

Section IV

Analytical and Control Aspects

Chapter 9

Surface Chemical Composition and Morphology

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9.1 OVERVIEW OF SURFACE COMPOSITION AND MORPHOLOGY

The chemical state in which a surface is left subsequent to a cleaning process is as important as the cleaning process itself. A surface that becomes recontaminated before the next processing step will not be useful. The best cleaning techniques are therefore the ones that can chemically passivate the semiconductor surface during the act of cleaning. It is essential to control the surface of silicon during the critical oxidation steps and epitaxial deposition steps.

The body of this chapter is broken up into four parts. The passivation of silicon surfaces is broken up into two parts, first the Si surfaces produced by cleaning processes that involve chemical oxidation (hydrophilic surfaces) are discussed in [Section 9.2](#) and then the surfaces produced by hydrofluoric acid (HF) etching (hydrophobic surfaces) are discussed in [Section 9.3](#). Contamination issues and chemical stability are discussed in [Sections 9.4 and 9.5](#).

9.2 SILICON OXIDE—TERMINATED SURFACES

9.2.1 Overview of Oxidized Surfaces

There are two predominant ways to chemically clean and passivate Si surfaces. The first is to grow a thin layer of SiO₂ in the act of cleaning. This is best accomplished using acidic or basic solutions mixed with H₂O₂ and is the basis of the RCA Standard Clean developed by Werner Kern in 1965 [1]. These clean processes leave 6–15 Å of hydroxylated oxide on the Si surface, which prevents recontamination of the Si. Such surfaces are hydrophilic in nature and are easily wetted by aqueous solutions. The second way to clean and passivate the surface is to completely dissolve the surface oxide in HF acid. Indeed, in 1958, the early electronic measurements of Buck and McKim [2] demonstrated the high degree of passivation of HF-treated Si surfaces. These surfaces now are known to be oxide-free and passivated with H. The H-terminated surfaces are hydrophobic in nature and not wetted by aqueous solutions.

The chemical composition of the Si surface subsequent to a clean is fundamental to its passivation. The chemically grown oxides exhibit a more complex chemical composition than the high-quality stoichiometric SiO_2 that is grown thermally. Thermal oxides have been studied extensively because of their application as gate insulators in the metal oxide semiconductor (MOS) technology (Section 9.2.1). However, as gate oxide thickness decreases, for sub-nanometer gate oxides, for example, the importance of the cleaning procedures employed is increasing, motivating in-depth studies of chemically grown oxides. The chemical oxides grown during preclean make up a large fraction of the total thickness and influence the overall performance of the dielectric. In addition, the deposition of high- k materials (k is the dielectric constant) by atomic layer deposition (ALD) can be influenced by the nature of the chemical oxide growth. The importance of understanding these chemical oxides cannot be overstated. They provide the foundation of the gate oxides and high- k dielectrics used today and of the future. Section 9.2.2 presents what is understood about the chemical composition of these hydrophilic surfaces.

The chemical composition of HF-etched Si surfaces was the subject of some confusion. It was originally believed that the hydrophobic nature of these surfaces was explained by F-termination and that the surfaces were hydrophobic because they resembled Teflon. It has since been shown that these surfaces are terminated with hydrogen and are therefore more paraffinlike because of the nonpolar nature of the Si–H bond. Because of this original confusion, some history of the evidence for H-termination is presented in Section 9.2.1. A chemical mechanism, which explains how the Si surfaces become H-terminated, is described.

The microstructural state of the surface has been shown to affect subsequent device properties [3–6]. Surface roughness can cause degradation of the breakdown field strengths of thin gate oxides [4–7], leading to decreased channel mobility [3–6]. Both hydrophilic and hydrophobic surface cleaning processes can affect the morphology of the Si surface and will be discussed separately in Section 9.2.2. Chemically grown oxides are noncrystalline, limiting the information obtained from most techniques for structural analysis. In contrast, HF etching leads to H-termination of the bulk crystal and can be studied in greater detail. To understand the variety of structures produced by different HF solution chemistries, the principles and limitations of the most useful structural analysis techniques briefly are reviewed in Section 9.2.3. Differences observed between the surface structures of Si (100) and Si (111) wafers after HF etching are then discussed. In particular, the structural morphology on both surfaces varies with solution pH, favoring the formation of (111) facets at high pH. The mechanism of this preferential etching is presented in Section 9.3.4.

Contamination is also an important issue for any cleaning or passivation process because trace amounts of impurities can drastically influence subsequent material properties. Contaminants can be intrinsic (e.g., H, OH, H_2O ,

O₂, S, Cl, and F) or extrinsic (e.g., C, Fe, Ni, Cr, and Cu) to the solutions used and will be treated separately. Contamination issues for oxide-covered surfaces are discussed in [Section 9.2.1](#). The contamination and the resulting loss of passivation of H-terminated surfaces are discussed in [Section 9.2.2](#).

As early as the 1950s, Atalla, Tannenbaum, and Scheibner [8] recognized the significance of oxide passivation of Si semiconductor devices. Their discovery of the unique passivation properties of thermally grown oxides, which led to remarkable improvements in device performance, is a cornerstone of modern Si technology. It is noteworthy that the importance of surface chemical cleaning before oxidation was already being stressed at the time of publication. In particular, the distinction between cleaning processes that resulted in hydrophilic versus hydrophobic surfaces was noted, but not understood at that time. Although this chapter focuses on chemically grown oxides formed during the cleaning process, many analogies to thermally grown oxides exist. Therefore, a brief discussion of thermal oxides is useful to an overall understanding of the different properties of thermally and chemically grown oxides.

Thermally grown oxides have been extremely well characterized because of their technological significance. Entire books have been devoted to this subject [9,10]; hence, only a brief summary follows. Heating Si wafers in either an O₂ or a H₂O environment forms thermally grown oxides. The resulting oxide is a noncrystalline, stoichiometric, and low-defect density form of SiO₂. This SiO₂ layer is largely impervious to contamination and protects the underlying Si substrate. A high degree of chemical passivation of the Si/SiO₂ interface is observed when measuring the electronic properties of the interface; for example, dangling bond densities below $1 \times 10^{12} \text{ cm}^{-2}$ are routinely achieved at the Si/SiO₂ interface ($1 \times 10^{11} \text{ cm}^{-2}$ subsequent to H₂ annealing) with fixed charge densities on the order of $1 \times 10^{11} \text{ cm}^{-2}$. The electronic perfection of this interface is the basis of the MOS technology and may be related to the atomic order of the Si/SiO₂ boundary.

Although the chemical perfection of the Si/SiO₂ interface is well established, the molecular structure is less well understood. It is clear that there is a transition region between the bulk stoichiometric amorphous SiO₂ and the crystalline Si substrate of 3–7 Å, where suboxides are observed by X-ray photoelectron spectroscopy (XPS) [11]. It is less clear, however, what the exact structure and composition are for this transition region. The various models proposed have been reviewed by Ourmazd and Bevk [12] and fall into three classes: (1) an epitaxial SiO₂ intervening layer, (2) a disordered substoichiometric oxide layer, and (3) an abrupt transition directly from the crystalline Si to the continuous random network of amorphous SiO₂. Although the exact structure of the Si/SiO₂ interface is still being debated, the transmission electron micrographs of Ourmazd and Bevk [12] demonstrate that atomically flat interfaces can be obtained from starting material with atomically flat Si (100) surfaces.

Chemically grown oxides are produced by various cleaning techniques that are based on the use of acidic and/or basic H_2O_2 aqueous solutions. The most widely used system is discussed in-depth in Chapters 1 and 4, the RCA Standard Clean developed by Kern and Puotinen [13]; a sequential two-step cleaning process where wafers are immersed first in a 1:1:5 solution of $\text{NH}_4\text{OH}:\text{H}_2\text{O}_2:\text{H}_2\text{O}$ at 80°C (SC-1) and then in a 1:1:5 solution of $\text{HCl}:\text{H}_2\text{O}_2:\text{H}_2\text{O}$ at 80°C (SC-2). The SPM (sulfuric acid–hydrogen peroxide mixture), also known as Piranha, consists of a 4:1 to 8:1 $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$ solution at a temperature in excess of 140°C . Other cleaning processes involve exposure to chemicals or solutions containing dissolved ozone (O_3). The oxides left behind after these surface treatments are similar to the thermally grown oxides in some respects but are quite different in others. The discussion of these chemically grown oxides is divided up into three parts: chemical composition (Section 9.2.2), structural morphology (Section 9.2.3), and contamination issues (Section 9.2.4).

9.2.2 Chemical Composition of Oxidized Surfaces

The properties of chemically grown oxides produced by the various integrated circuit (IC) cleaning techniques are quite similar and have been addressed by Deal et al. [14]. These techniques are covered in Chapter 4 in detail; they include SPM, SC-1 (standard clean 1), and SC-2 (standard clean 2). The oxides tend to be $\sim 10\text{--}15\text{ \AA}$ thick, depending on the process temperature and the solution chemistry used [15]. These films are largely stoichiometric but, because they are so thin, exhibit properties characteristic of the interfacial transition regions of thicker, thermally grown oxides. The large suboxide content that is characteristic of these chemically grown oxides is shown in the Si 2p core-level spectra performed by Sugiyama et al. [16] in Fig. 9.2-1. For the two different chemical preparations shown, the spectra are dominated by the Si crystal substrate and by stoichiometric SiO_2 , Si^{4+} . There is, however, a relatively strong and unambiguous Si^{2+} contribution corresponding to an interfacial transition region similar to that observed for thermal oxides. In this case, the Si^{2+} contribution is 10%–20% of the Si^{4+} contribution, indicating that the transition region is a large fraction of the surface layer. In addition to Si^{2+} , also observed are varying degrees of other suboxides that depend on the exact surface treatment performed [15,16], but these will not be discussed here. Dangling bond defects at the Si/SiO₂ interface have been quantified using minority carrier lifetime measurements to extract the surface recombination velocity and surface defect density [17,18]. Defect densities on the order of $1 \times 10^{13}\text{ cm}^{-2}$ are typical for these clean native oxides, placing these surfaces at a level 2 orders of magnitude higher defect density than for the thermal oxides.

Chemical oxidation involves species other than Si and O. Aqueous solutions are predominantly composed of H_2O and are, therefore, sources of H_2O ,

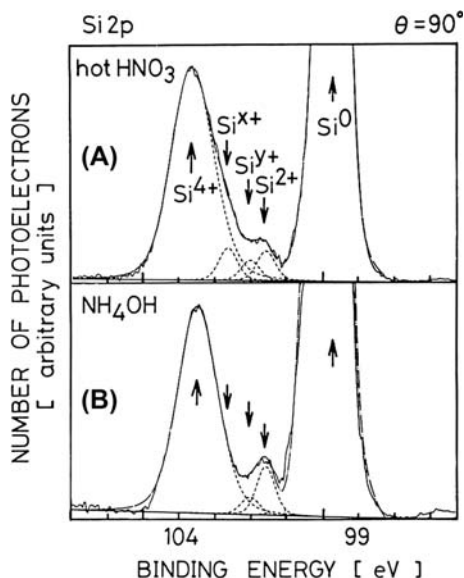


FIGURE 9.2-1 X-ray photoelectron spectra of the Si $2p_{3/2}$ core level associated with native oxides on Si formed by immersion in (A) HNO_3 at $45\text{--}50^\circ\text{C}$ for 5 min, and (B) $1:1:4 \text{ NH}_4\text{OH}:\text{H}_2\text{O}_2:\text{H}_2\text{O}$ at $63\text{--}80^\circ\text{C}$ for 10 min. A thorough rinse in DIW for 10 min is performed in both cases before introduction into the ultrahigh vacuum chamber. The spin orbit splitting of the Si $2p$ level is removed, the data smoothed and the background subtracted. The *dashed lines* are the result of a spectral deconvolution performed by assuming that the chemical shifts and the values of full-width half-maximum for the various components are the same as for the Si^{4+} , which is associated with stoichiometric SiO_2 [16]. Reprinted with permission from K. Sugiyama, T. Igarashi, K. Moriki, Y. Nagasawa, T. Aoyama, R. Sugino, T. Ito, T. Hattori, *Jpn. J. Appl. Phys. Part 2* 29 (12) (1990) L2401.

OH, and H. Thus, these chemical species are incorporated into the oxide layer to some extent. There is extensive literature on the infrared (IR) spectroscopy of OH and H_2O in and on the surfaces of silica glasses [19]. These OH units are found in the form of Si-O-H or H-O-H . When the silica glasses are treated at elevated temperatures (750°C) under vacuum, a narrow (50 cm^{-1}) IR absorption band is observed at 3750 cm^{-1} . This absorption is assigned to the O-H stretching vibration and is characteristic of isolated Si-O-H units. Without such thermal treatment, the IR absorption of the O-H stretch is extremely broad ($\sim 400 \text{ cm}^{-1}$) and peaks near 3400 cm^{-1} , indicating a strong interaction between neighboring OH and H_2O groups. This interaction, commonly referred to as H bonding, is simply a weak bonding between an H atom and an O atom of neighboring molecules or molecular complexes. The exact nature of this bond is not completely understood. It is partially ionic and partially covalent and weakens the O-H bond, causing inhomogeneous broadening on the low-frequency side of the isolated O-H stretching vibration.

Hydrogen bonding is common to all aqueous solutions and is intimately related to the heat of solvation and to the hydrophilic nature of oxide-covered Si surfaces. Hydrogen-bonded O—H is observed on all chemically grown oxides. An example of an IR absorption spectrum of such a surface is shown in Fig. 9.2-2, where an H-terminated Si surface is chemically oxidized using the SC-2 step of the RCA clean. The spectra presented in Fig. 9.2-2 are ratios of oxidized and H-terminated Si surfaces and, therefore, display a negative absorption for the Si—H stretch bands (2100 cm^{-1}) associated with the reference surface. The H-bonded OH stretching vibration peak is at $\sim 3300\text{ cm}^{-1}$ and is $\sim 400\text{ cm}^{-1}$ wide (full-width half-maximum) with an asymmetric line shape. It is quite difficult to distinguish between Si—O—H and H_2O in these spectra without additional spectral information from the scissor mode of the H_2O molecule in the $1600\text{--}1700\text{ cm}^{-1}$ region. The intensity difference observed between the spectra taken with s- and p-polarization indicates that the OH groups must reside in or on the oxide layer. Although it is unclear from these spectra, in very general terms, most of the IR signal comes from H_2O on the surface of the oxide, because gentle heating (100°C) substantially decreases the OH/ H_2O absorption. The surface OH concentration is extremely important in determining the initial reaction rate for high-k ALD reactions requiring surface OH to promote reactions [20]. It is of interest to note that OH content can be varied depending on the chemical treatment chosen. Ozone (O_3) treatments are perhaps the most oxidizing and produce the

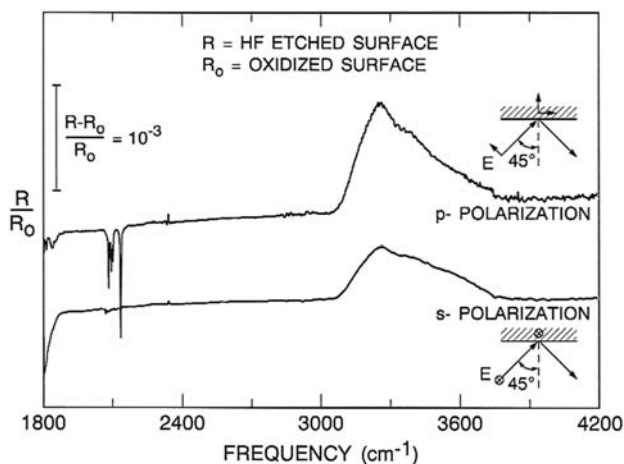


FIGURE 9.2-2 Infrared absorption spectra of a Si wafer chemically oxidized in a 1:1:4 solution of $\text{HCl}:\text{H}_2\text{O}_2:\text{H}_2\text{O}$ at 80°C for 10 min. The ratio of the spectra lines were compared with the corresponding spectra of the H-terminated Si prepared by etching in a buffered HF solution. A multiple internal reflection geometry was used with 75 reflections at a 45-degree internal angle of incidence, as shown in the inset [95]. Reprinted with permission from P. Jakob, Y. Chabal, *J. Chem. Phys.* 95 (1991) 2897. Copyright 1991, American Institute of Physics.

lowest amount of OH on the surface. SC-1 solutions, on the other hand, produce larger quantities of OH units on the surface and within the films. It appears that the OH concentration is related to the degree in which the cleaning solutions produce SiO₂. More OH-terminated groups are formed when the oxide formation is less complete and vice versa.

It is also apparent that Si—H units reside on Si surfaces upon which chemical oxides have been grown [15,16]. The first convincing evidence came from the IR spectra of Ogawa et al. [15] shown in Fig. 9.2-3, where Si—H stretching vibrations are identified at $\sim 2260\text{ cm}^{-1}$. Si—H stretches in that region of the spectrum originate from Si—H where the Si atom is back-bonded to O atoms [21]. This evidence indicates that the Si—H resides within the oxide matrix. The area density is estimated to be $(2\text{--}3) \times 10^{13}\text{ cm}^{-2}$. XPS

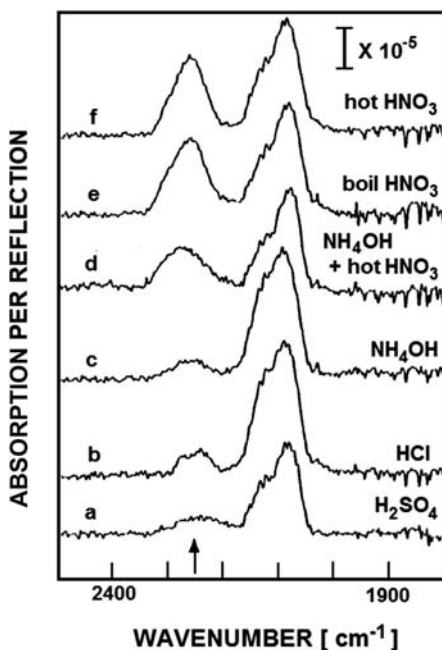


FIGURE 9.2-3 Infrared absorption spectra of six different native oxides on Si wafers: (A) “H₂SO₄” corresponds to a 10 min treatment in 4:1 H₂SO₄:H₂O₂ at 85–90°C, (B) “HCl” corresponds to 10 min in 1:1:4 HCl:H₂O₂:H₂O at 37–65°C, (C) “NH₄OH” corresponds to 10 min in 1:1:4 NH₄OH:H₂O₂:H₂O at 63–80°C, (D) “NH₄OH + hot HNO₃” corresponds to “NH₄OH” followed by “hot HNO₃” (E) “boil HNO₃” corresponds to treatment in HNO₃ at 115–125°C, and (F) “hot HNO₃” corresponds to 5 min in HNO₃ at 45–60°C. The absorption (indicated by the arrow) which peaks at $\sim 2260\text{ cm}^{-1}$ arises from Si—H stretches where the Si is back-bonded to O (i.e., Si—H in or on the SiO₂) [15]. Reprinted from H. Ogawa, N. Terada, K. Sugiyama, K. Moriki, N. Miyata, T. Aoyama, R. Sugino, T. Ito, T. Hattori, *Silicon -hydrogen-Bonds in silicon oxide near the SiO₂/Si Interface*, *Appl. Surf. Sci.* 56–58 (1992) 836. Copyright (1992), with permission from Elsevier.

data from the same group suggest that these Si—H units are actually localized near the surface of the oxide. If this is indeed the case, these units may be residual Si—H bonds from the original H-terminated hydrophobic surface before the chemical oxidation. This agrees well with the theory that oxidation proceeds via O-atom insertion between the Si—Si back-bonds of the surface and is consistent with observations on the initial stages of oxidation of hydrophobic surfaces [22,23]. Also observed in the spectra of Fig. 9.2-3 are Si—H stretches in the range of 2080 cm^{-1} , which are best explained by H-atoms bonded to substrate Si (i.e., Si atoms that are back-bonded to Si atoms rather than to O atoms). The high-frequency shoulder of this band ($\sim 2140\text{ cm}^{-1}$) is most likely associated with Si—H stretches where some of the Si back-bonds are attached to an O atom. If the mode at 2080 cm^{-1} does arise from Si—H at the Si/SiO₂ interface, it is important to determine the formation mechanism. Limited research has been performed to understand defect generation and their effects at the Si—SiO₂ interface—[24,25].

9.2.3 Structures and Morphology of Oxidized Surfaces

The work of Hahn and Henzler [26], Heyns et al. [7], and Ohmi et al. [5] were some of the first to correlate electronic device properties with surface structural properties. Although it is intuitively obvious that surface roughness must be detrimental to semiconductor devices at some scale, the main contribution of these works has been to define at what scale surface roughness is important to device yield and reliability. The evidence for degradation of thin gate oxide (30–50 Å) breakdown field strength and channel mobility with surface roughness on a microscopic scale has been realized early by the IC industry [27]. This issue has been partially resolved by incorporating an ultrathin SiO₂ layer at the interface between high-k dielectrics and silicon, with special attention to surface roughness [28,29].

The morphology that a surface exhibits tends to be a function of the complete processing history experienced by the wafer, and it is therefore quite complicated. The initial surface polish processes, chemical cleaning processes, thermal oxidation processes, and etching processes, all influence the surface morphology. This section begins with a discussion of substrate wafers, including those prepared via chemical mechanical polishing (CMP) and epitaxy. It then considers the effects of chemical cleaning on surface morphology. Finally, future trends in controlling oxidation and interfacial structure are briefly discussed.

Near atomic perfection is achieved through surface CMP. Commercial wafers exhibit a typical surface roughness on the order of 2 Å RMS (root mean square) and surface finishes produced in the laboratory have approached 1-Å RMS [30]. A scanning tunneling microscope (STM) image of such a surface is shown in Fig. 9.2-4 [31]. Although STM images can characterize surface roughness on length scales covering the range from 1.0 to 1000 Å, these

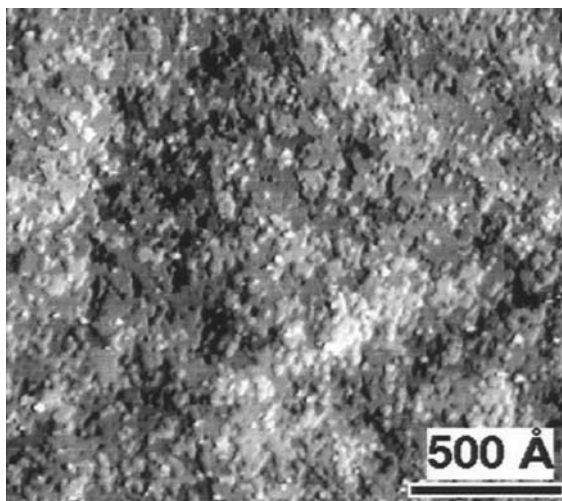


FIGURE 9.2-4 Scanning tunneling microscope image of a polished Si (111) surface exhibiting $\sim 1.6\text{-}\text{\AA}$ RMS roughness. The image was taken immediately following a buffered HF (7:1 HF:NH₄F) immersion [31]. Reprinted from T.A. Newton, J.A. Boiani, M.A. Hines, *The correlation between surface morphology and spectral lineshape: a re-examination of the H–Si (111) stretch vibration*, *Surf. Sci.* 430 (1999) 67. Copyright (1999), with permission from Elsevier.

surfaces also are characterized with a variety of other techniques, such as diffraction methods, which span lengths up to 1 mm, with good correlation observed on all scales [30]. The post-CMP STM image shown in Fig. 9.2-4 is taken subsequent to HF removal of surface oxidation, which renders the surface hydrophobic and H-terminated. Similar surface roughness is found for Si (100) and Si (111) surfaces. The process to produce H-terminated surfaces briefly will be discussed in Section 9.2.2.

Hydrogen-terminated surfaces are not as stable as oxide-terminated surfaces, and thus it is not surprising that wafer vendors ship wafers in the hydrophilic (oxide covered) state. The wafer manufacturer's polish and clean recipes are proprietary, but the wafers receive an RCA clean and other cleaning and polishing steps before they are shipped. Another technique that provides atomically perfect surfaces is Si molecular beam epitaxy [32]; however, this technique is expensive for widespread commercial implementation. Surfaces formed during commercial Si epitaxial growth by chemical vapor deposition, however, may also be made atomically perfect under the proper conditions. As-received Czochralski Si and epitaxial Si substrates both exhibit a surface roughness of $\sim 2\text{-}\text{\AA}$ RMS.

The consensus is that the acidic peroxide cleaning processes (SPM or SC-2) do not cause a substantial increase in the microscopic roughness of as-received wafers. The SC-1 cleaning process (typically 1:1:5 NH₄OH:-H₂O₂:H₂O at 80°C), on the other hand, has been found to substantially

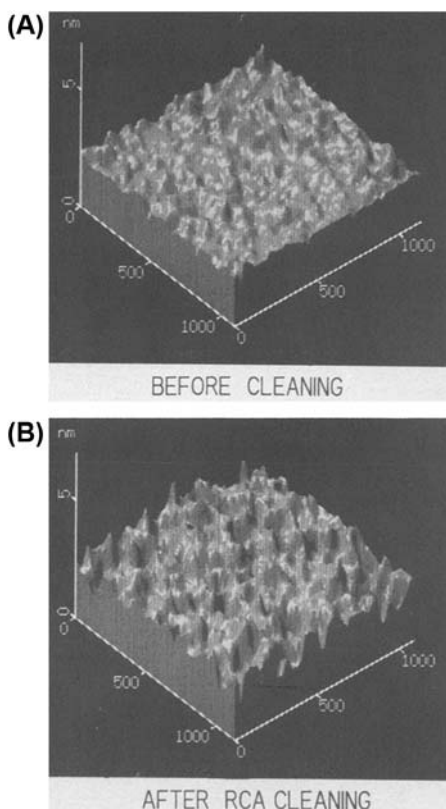


FIGURE 9.2-5 Scanning tunneling microscope images of Si (100) surfaces taken (A) before and (B) after an RCA Standard Clean (SC-1 and SC-2) [4]. Reproduced by permission of ECS—The Electrochemical Society from M. Miyashita, T. Tusga, K. Makiyara, T. Ohmi, *J. Electrochem. Soc.* 139 (3) (1992) 2133.

increase the surface roughness [33]. A comparison of the surface roughness of a control wafer and a wafer cleaned in a standard SC-1 process is shown in the STM images of Fig. 9.2-5. Control wafers exhibit an RMS roughness of 2 Å. The SC-1 treatment more than doubles the observed roughness and repetitive SC-1 cycles can increase it by as much as a factor of 5, approaching 10-Å RMS. This roughness has been shown to decrease breakdown field strengths by as much as 30% [6,34] and to degrade channel mobility by factors of two to three times [5].

The mechanism leading to the roughening in the basic peroxide solution is not completely understood but is related to the slow but finite Si etching rate in the SC-1 solution (~ 8 Å/min at full strength at 80°C) [6]. The acidic peroxide solutions, on the other hand, do not etch and hence do not roughen.

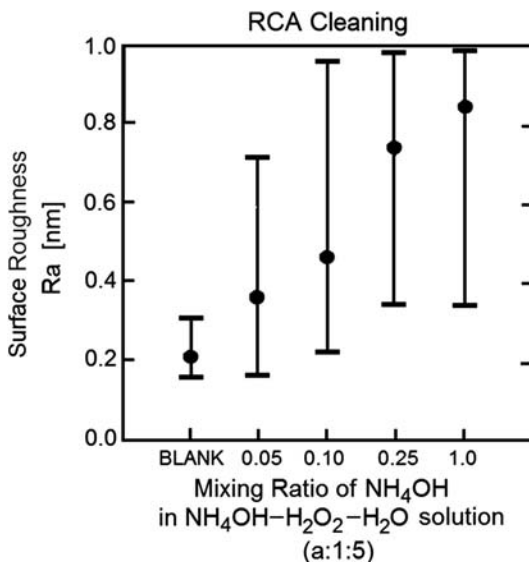


FIGURE 9.2-6 Surface roughness plotted as a function of NH_4OH concentration in a 10-min $\text{NH}_4\text{OH}:\text{H}_2\text{O}_2:\text{H}_2\text{O}$ solution treatment at 85°C [6]. T. Ohmi, M. Miyashita, M. Itano, T. Imaoka, I. Kawanabe, *Dependence of thin-oxide films quality on surface microroughness*, *IEEE Trans. Electron Devices*, 39 (3) (1992) 537. Copyright (1992) IEEE. Used with permission.

A proposed solution to the basic peroxide roughening problem is to reduce the etching rate by reducing the concentration of NH_4OH in the SC-1 solution [33]. The etching rate drops to $1 \text{ \AA}/\text{min}$ if the concentration of NH_4OH is decreased by a factor of 100 times. Fig. 9.2-6 shows a plot of the measured RMS roughness as a function of NH_4OH concentration [6]. The industry, however, is working hard to keep the standard SC-1 solution even though, in some case, it is detrimental to the surfaces. The reason is simple: SC-1 is one of the most efficient particle removal agents known. Further, the fact that the basic peroxide solution slightly etches both SiO_2 and Si is precisely why it is such an efficient particle remover. Chapter 4 discusses the development of a low-concentration, high-dilution alkaline solution that achieved the desired particle removal efficiency, cleanliness, and roughness.

The microscopic mechanism by which the etching roughens the surface also has been investigated. It should be mentioned that etching alone does not necessarily mean the surface roughness will increase. It is nonuniform etching or preferential etching around a defect or contaminant, which is the true culprit. An early example is the work of Verhaverbeke et al. [35] who found that the Ca concentration in SC-1 dramatically changes the degree to which the surface roughens.

The exact molecular structure of these chemically grown Si/ SiO_2 interfaces is very difficult to deduce. Native oxide growth has been shown to occur in an

extremely controlled manner [32,36], leading to atomically ordered interfaces under the right conditions. This phenomenon appears, in fact, to be more general. In very careful XPS studies of surface O concentration as a function of time, layer-by-layer oxidation of Si has been observed [37,38]. Layer-by-layer oxidation, of course, requires that the previous layer finish before the next layer begins oxidation and leads by necessity to the conclusion that some form of order must exist at the Si/SiO₂ interface.

9.2.4 Contamination Issues Associated With Oxidized Surfaces

One of the main objectives for the development of the RCA clean was to remove organic and metal contaminants from the surface of Si wafers [1,13]. Although the RCA clean was developed over 40 years ago, it has functioned extremely well and is still the dominant clean used before gate oxidation. As gate oxides move into the sub-100 Å regime, there has been renewed interest in how trace metal contamination in the pregate cleaning processes affects gate oxide properties [7]. For instance, Pan et al. [39] looked at the effect of different metal contaminants on breakdown strength and leakage current and found that the electrical breakdown voltage decreases with increased metallic contamination, partly because the effective thermal oxide thickness decreases and partly because of increased electrical defect states. Residual trace metal contamination at the $1 \times 10^8 \text{ cm}^{-2}$ level is observed after RCA cleaning and is dependent on the quality of the chemicals used. Contamination levels required for device manufacturing are discussed in detail in Chapter 2. The presence of these metals can lead to surface roughening because of a number of different mechanisms that are still being debated. In addition to preferential etching mechanisms, it has been suggested that metals can enhance oxidation rates and therefore increase surface etching rates nonuniformly leading to surface roughening. Another mechanism is related to bubble formation that blocks surface reactions, again leading to increases in surface topography. Additionally, metals can get trapped in the oxide formed during the SC-1/SC-2 cleaning process and can subsequently lead to leaky junctions and yield and reliability problems in gate oxides [7]. Ozone oxidation, sometimes with the addition of HCl, has previously been used to grow passivating chemical oxides and avoid metal contamination from the RCA chemicals. However, ozonated H₂O also can be produced in extremely high purity and is more commonly used currently to obtain good quality gate dielectrics [40,41].

Another common contaminant on these oxide-covered surfaces is carbon. Carbon is most likely incorporated in or on these surfaces in the form of hydrocarbons and can come from the chemicals, the H₂O used to rinse the wafers, or from the air in the laboratory environment because of material outgassing [42,43]. Trace hydrocarbons, such as those deposited on the silicon surface from airborne molecular contaminants (AMCs), have proven to be detrimental to gate oxide integrity [44,45] and much work is being done to

mitigate their presence [46]. Additionally, if handled improperly, the AMC can turn into SiC precipitates which can cause weak spots in the oxides grown [47].

Hydrocarbon contamination is also a concern for surface preparation before the epitaxial growth of Si. In this case, growing defect-free Si surfaces that are completely free of contamination are needed and therefore C contamination is of critical concern. The technique of desorbing the oxide at elevated temperature before epitaxy was first discussed by Henderson [48] in which the results showed that atomically clean surfaces with only a small amount of C residue could be obtained after the RCA Clean. Ishizaka and Shiraki [49] reduced the level of C entrained in the oxide by repetitively immersing the wafers in boiling HNO₃ acid followed by HF, ending with a concentrated SC-2 type of clean (1:1:4 H₂O:H₂O₂:HCl at 90–100°C). Another efficient technique to remove hydrocarbons is exposure to UV/O₃ (ultraviolet/ozone) [50] and is discussed in detail in Chapter 6.

Some contaminants, such as S and Cl, can originate directly from the solution used during the chemical oxidation. Sulfur and chlorine have been observed from the SPM and SC-2 cleaning solutions, respectively (see Chapters 2 and 4). Fluorine, on the other hand, has been observed when the chemical oxidation is preceded by an HF treatment. In this case, F is found to segregate at the Si/SiO₂ interface [51].

9.3 HYDROGEN-TERMINATED SILICON SURFACES

9.3.1 Chemical Composition of HF-Treated Surfaces

The unique properties of Si surfaces treated in HF solutions were recognized in the 1950s; however, insight into the chemistry and structures of these hydrophobic surfaces only began in the 1980s and continues to this day [52–54]. It is now clear that H- and not F-termination of the dangling bonds on Si surfaces explained the hydrophobicity, the high resistance to chemical attack, and the low surface recombination velocity. Fluorine can be found on these surfaces, but only in small quantities and should be thought of as a minor contaminant rather than a major constituent of the surface. The initial confusion and controversy about H- versus F-termination was quite understandable. Most conventional surface spectroscopic methods (e.g., AES—Auger electron spectroscopy, XPS) rely on core electrons and cannot measure H, because it has no core. These methods are instead sensitive to species such as F, which is detected to some degree. Vibration spectroscopies (e.g., EELS—electron energy loss spectroscopy, IR), on the other hand, are extremely sensitive to H-containing surface species but were not widespread in the wet chemistry community until the 1980s. This initial controversy is a good example of how processing issues can be addressed by the surface science community and therefore it is worthwhile to give a brief historical background. Chapter 13 reviews various analytical methods for detecting contamination.

Although several early reports of H-termination subsequent to HF etching of Si surfaces emerged in the time period between 1965 and 1975 [55,56], it was not until 1984 that Ubara, Imura, and Hiraki [57] showed that the removal of SiO_2 in HF solutions results in the formation of SiH on the etched Si surface. Their results obtained on hydrogenated microcrystalline Si [58,59] are summarized in Fig. 9.3-7. The (A) as-grown samples are (B) thermally oxidized at low temperature, then (C) HF-solution etched, (D) reoxidized at much higher temperature, and finally (E) HF-solution etched again. The main result is best summarized in curves (D) and (E). After high-temperature thermal oxidation in air at 600°C , all traces of H have disappeared, shown in curve (D), as evidenced by the absence of the SiH_x bands centered at 2100 cm^{-1} . This treatment produces strong Si-O and Si-O_2 bands centered at 1100 cm^{-1} , confirming the formation of an oxide layer. Subsequent dipping in HF solution eliminates all the bands associated with Si-O and gives a

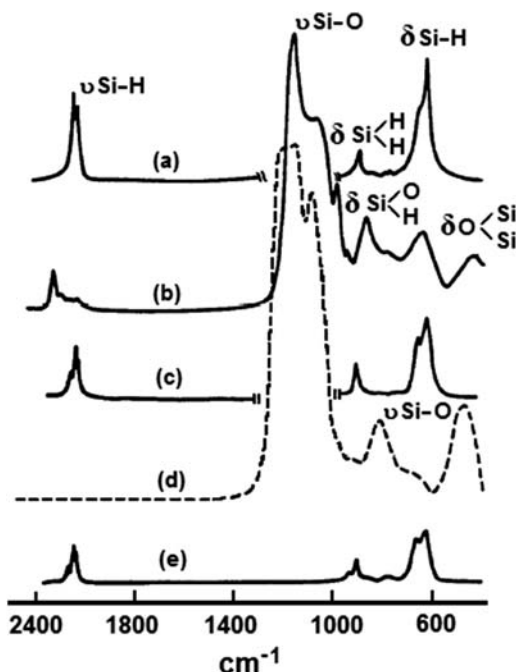


FIGURE 9.3-7 Transmission infrared absorption spectra of hydrogenated microcrystalline silicon ($\mu\text{c-Si:H}$): (A) as grown using radio-frequency reactive sputtering in H_2 at 250°C , (B) after thermal oxidation in air at 200°C for 5 h, (C) after subsequent HF etching of the previously oxidized surface, (D) after a thermal oxidation at 600°C for 1 h, and (E) after subsequent HF etching of this previously thermally oxidized surface [57]. Reprinted from H. Ubara, T. Imura, A. Hiraki, *Formation of Si-H bonds on the surface of microcrystalline silicon covered with SiO_x by HF treatment*, *Sol. Stat. Commun.* 50 (7) (1984) 673. Copyright (1984), with permission from Elsevier.

spectrum with prominent Si—H bands at 2100 cm^{-1} (SiH_x stretch), 900 cm^{-1} (SiH_2 scissor), and 650 cm^{-1} (SiH_x bend), as shown in curve (E). These data show the removal of SiO_2 in HF with subsequent termination of the surface by H. Ubara et al. [57] postulated a mechanism to account for the H-termination, which was later confirmed by *ab initio* calculations [60] and will be discussed in detail later in this section.

Despite these early works, which clearly pointed to hydride formation upon HF solution etching, it was still widely believed that F was in fact the passivating agent. Raider et al. [61] attributed the hydrophobic nature of HF-etched surfaces to the presence of Si—F bonds or adsorbed HF. Licciardello et al. [62] indicated that the presence of an organic (hydrophobic) overlayer and not the Si—F surface per se was responsible for the hydrophobicity, although they still assumed that this overlayer was deposited on top of the F-terminated Si surface. In addition, Weinberger et al. [63] explained the electronic passivation of HF-treated Si wafers as being because of F-termination of the Si dangling bonds. They further argued that the strength of the Si—F bond would make it an extremely stable surface, a point discussed later in this section. Indeed, indirect support for F passivation was drawn from the observation that F is a stable adsorbate on the surfaces of Si prepared in ultrahigh vacuum (UHV) [64]. Furthermore, F was directly identified by XPS on HF-treated Si surfaces [53], suggesting that a monolayer (ML) of F is present on the surface. Early work using radioactive ^{18}F showed that F ions were indeed adsorbed on the surface [65].

The year 1986 was perhaps the turning point in the H- versus F-termination debate. Yablonovitch et al. [18] observed that the F concentration, as measured by XPS, was highly variable and could not be the correct explanation for the remarkable surface passivation achieved subsequent to HF etching. Using IR spectroscopy, they showed that a ML of H was adsorbed on the surface of HF-etched Si, with a spectrum characteristic of clean Si—H bonds (i.e., without O or F as a nearest neighbor). Another clear demonstration of the variability of the F concentration on the surface came from the XPS data of Grunthaner and Grunthaner [66] showing no detectable O, F, N, or S on HF/ethanol spin-etched Si samples. Fig. 9.3-8, for instance, shows the Si 2p core-level spectra measured after different chemical treatments. The spin-etched samples are characterized by an unshifted Si 2p core level, making it possible to set an upper limit of 0.1% of an ML of Si directly bonded to electronegative elements, such as F or O, for the spin-etched samples [67]. Independently, Grundner and Jacob [68] published the results of detailed EELS and XPS studies of oxidized and HF-etched Si surfaces. The EELS data, shown in Fig. 9.3-9, consistently show strong H vibrations at 2100 , 900 , and 650 cm^{-1} , although hydrocarbons (2800 cm^{-1}) and OH (3400 cm^{-1}) vibrations also contribute to the spectra [68–70]. The intensity of the F 1s line measured in XPS data [69,70] indicates that the F concentration, after a DI (deionized) H_2O (DIW) rinse, is always less than 1%–2% ML. This work

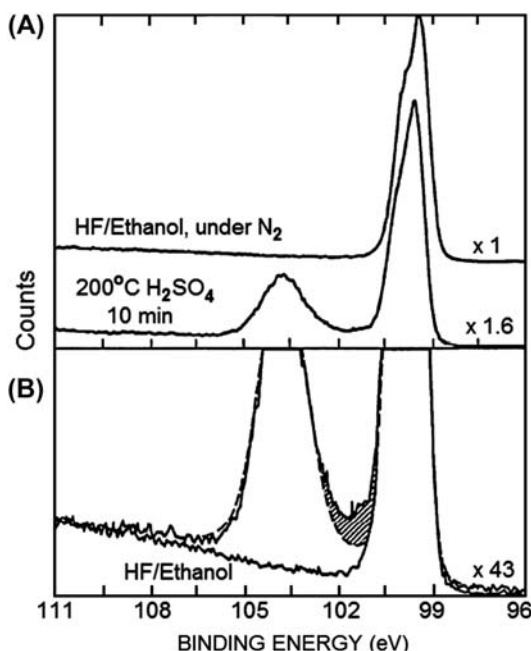


FIGURE 9.3-8 (A) Typical X-ray photoelectron spectra of the Si 2p core level obtained before (lower spectrum) and after (upper spectrum) a spin-etch under N_2 using HF in ethanol. (B) Expansion and overlay of the data shown in (A) to emphasize the region between the peaks that correspond to the Si substrate and to the SiO_2 . The dashed line represents a least-square fit to these two major components. The cross-hatched region corresponds to Si suboxide species [67]. Reprinted from P.J. Grunthaner, F.J. Grunthaner, R.W. Fathauer, T.L. Lin, M.H. Hecht, L.D. Bell, W.J. Kaiser, F.D. Schowengerdt, J.H. Mazur, *Hydrogen-terminated silicon substrates for low-temperature molecular beam epitaxy*, *Thin Solid Films* 183 (1989) 197. Copyright (1989), with permission from Elsevier.

confirms the formation of silicon hydrides upon HF etching and suggests that F is a contaminant that can be removed by rinsing in H_2O .

Both the surface-recombination velocity measurements of Yablonovitch et al. [18] and the extensive XPS and EELS studies of Grundner et al. [68–70] motivated a number of photoelectron emission and IR absorption studies [71–73]. Takahagi et al. [71] used XPS, UV photoemission spectroscopy and IR absorption to quantify the chemical species on the Si surface after a combination of UV/O_3 cleaning and HF dipping. Using dilute HF solutions, they achieved low levels of contamination (totaling less than 5% of a ML of O, C, and F) and detected both Si–H and Si– H_2 species on the Si (100) surface. A typical XPS survey scan is shown in Fig. 9.3-10, which emphasizes the chemical purity of HF-cleaned surfaces.

Burrows et al. [72] and Chabal et al. [73] investigated the Si–H stretch vibrations ($2000\text{--}2200\text{ cm}^{-1}$ region) to quantify the surface morphology of Si

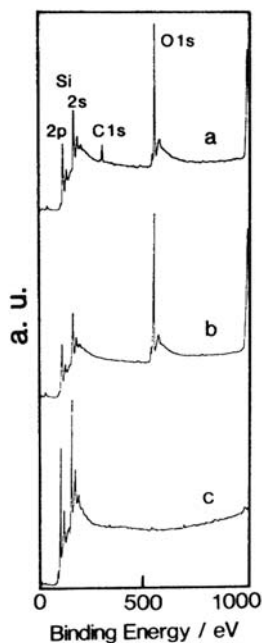


FIGURE 9.3-10 X-ray photoelectron spectroscopy survey scan spectra of Si surfaces (A) before UV/O₃ cleaning (as purchased with 7-Å thick native oxide layer and a 2-Å thick organic contamination layer), (B) after UV/O₃ cleaning with a low-pressure Hg lamp (184.9 and 253.7 nm emission) in an O₂ atmosphere, and (C) after subsequent HF dipping of the UV-cleaned sample in an 1 wt% HF solution [71]. Reprinted with permission from T. Takahagi, I. Nagai, A. Ishitani, H. Kuroda, Y. Nagasawa, *J. Appl. Phys.* 64 (1988) 3516. Copyright 1988, American Institute of Physics.

vibrations [75]. This is in contrast to the early EELS data [69,70] shown in Fig. 9.3-9, where relatively intense losses around 800–1100 cm⁻¹ (oxide) and 2900 cm⁻¹ (hydrocarbons) are apparent. Possible reasons for the discrepancy are variations in (1) the purity of the chemicals used, (2) the handling and rinsing procedures, and (3) the wafer introduction and evacuation procedures used for these UHV studies.

The main conclusion of the above studies is that H acts as the passivating agent and is the direct result of HF etching. The concentration of contaminants, such as C, O, and F, depends on the details of the processing, in particular, the rinsing procedure after the last etching step directly affects the concentrations of F and O.

9.3.2 Mechanism of Hydrogen Termination

Part of the confusion concerning F-termination of the Si following HF etching has arisen not only because of the stability of the Si–F bonds but also because

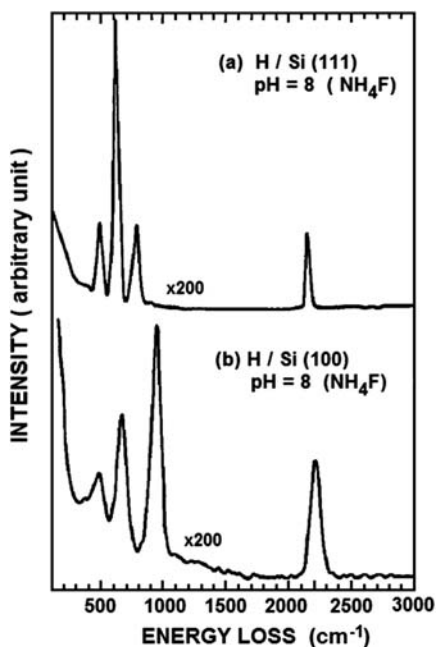
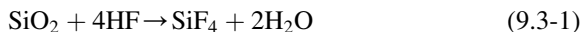


FIGURE 9.3-11 Electron energy loss spectra of (A) Si (111) and (B) Si (100) after etching in an ammonium fluoride solution ($\text{NH}_4\text{F}-\text{H}_2\text{O}$, $\text{pH} = 7.8$) and a brief (~ 10 s) rinsing in DIW [75]. The $\times 200$ factor corresponds to the magnification of the spectrum relative to the elastic peak. Adapted from P. Dumas, Y.J. Chabal, *Electron-energy-loss characterization of the H-terminated Si (111) and Si (100) surfaces obtained by etching in NH_4F* , *Chem. Phys. Lett.* 181 (1991) 537. Copyright 1991, with permission from Elsevier.

the accepted explanation of the mechanism for SiO_2 dissolution leads automatically to F-terminated Si. The dissolution of SiO_2 by HF can be depicted in its simplest form in the following reaction:



Notice that the above reaction involves HF molecules and not F^- ions in solution. HF is a weak acid having an equilibrium constant such that it does not dissociate readily in concentrated solutions [77,78]. In addition, Judge [77] showed that even if F^- ions are available, they give an etching rate, which is negligible compared with HF and HF_2^- and particularly H_2F_2 species [79,80]. Thus, HF and its associated forms need be considered in the dissolution mechanism. As an illustration only of the earlier thinking in the field, the attack of Si–O bonds by HF molecules is considered; this would occur by the insertion of HF between the Si–O atoms. This reaction is depicted schematically in Fig. 9.3-12A as if it were the last Si–O bond to be broken before reaching the Si substrate. This insertion occurs with a low activation barrier

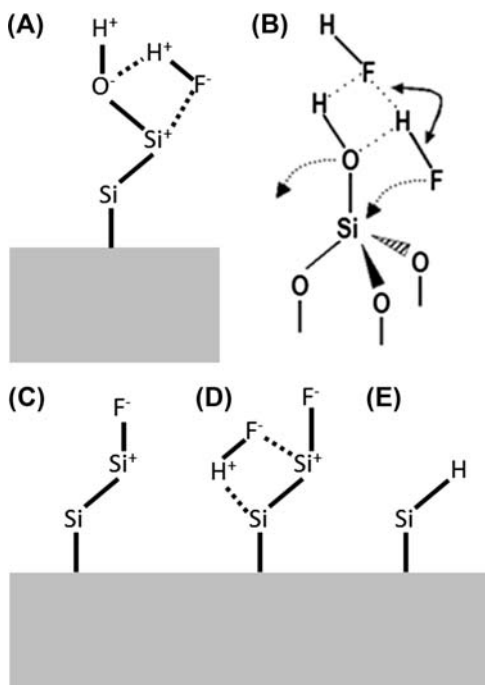


FIGURE 9.3-12 Schematic representation of Si etching and H passivation by HF or H₂F₂. Panel A shows the insertion of HF over the initial Si-OH termination while B shows the mechanism using H₂F₂ as the reactant [79]. Panels C–E illustrated the transformation from an initial Si–F termination to the final H-terminated Si surface. Panels A, C–E are used with permission from Y. J. Chabal. Panel B is reprinted from D.M. Knotter, *Etching Mechanism of Vitreous Silicon Dioxide in HF-Based Solutions*, *J. Am. Chem. Soc.* 122 (2000) 4345–4351. Copyright (2000) American Chemical Society.

because the reaction is highly exothermic and conserves the number of broken and reformed bonds. The reaction also is greatly facilitated by the highly polar nature of the Si–O bond, which the highly polar HF molecule can use to its advantage during attack. The Coulomb attraction naturally leads to having the positively charged H atom associated with the negatively charged O atom, and the negatively charged F atom associated with the positively charged Si atom of the Si–O bond. This liberates H₂O into the solution and leaves Si–F in its place on the surface (Fig. 9.3-12C). The Si–F bond is the strongest single bond known in chemistry with a bond energy of ~ 6 eV [81]. The bond strength of the Si–H is only ~ 3.5 eV and based on these thermodynamic considerations, the F-terminated surface must be more stable than the H-terminated surface. Nowadays, the general consensus of the industry is that H₂F₂ is the primary etchant of SiO₂ because it has a higher equilibrium constant in solution [79,80]. Here, it is proposed that the initial Si–F

formation is formed by consecutive elimination and addition reactions in which HF_2^- is inserted across the Si—OH bond liberating H_2O and F^- as shown in Fig. 9.3-12B [79].

Ubara et al. [57] were the first to propose a reaction mechanism to get around this dilemma. They recognized that the Si—F bond must be highly polar because of the large electronegativity difference between the atoms. They suggested that the Si—F bond causes bond polarization of the Si—Si back-bond allowing HF attack of the back-bond, as illustrated in Fig. 9.3-12D. This kinetically favorable pathway results in the release of stable SiF_x species into the solution leaving Si—H behind on the surface, shown in Fig. 9.3-12E. The validity of this proposed pathway was confirmed using first principles molecular orbital calculations of the activation energies of these types of reactions on model compounds by Trucks et al. [60]. In these calculations, an activation energy of ~ 1.0 eV was found for reactions shown in Fig. 9.3-12C. Ubara et al. suggested that the low activation energy is caused by charge transfer between the highly electronegative Si and F atoms [57]. In the absence of polarization, as is the case for nonpolar Si—H bonds, the activation energy of the Si—Si back-bond attack is 1.6 eV, which is 0.6 eV higher in energy than that for fluorinated Si species. The impact of the Coulomb interaction also can be observed by inverting the HF molecule, making the attack occur in opposition to the Coulomb force. In that case, an activation energy of 1.4 eV is obtained. In summary, HF attacks polar species very effectively, but is much less effective against nonpolar species [60]. Also, the reactant must attack the bonds in a specific orientation to take advantage of the Coulomb interaction between the positively and negatively charged atoms. These concepts allow us to understand why it is H and not F that terminates the Si dangling bonds after HF solution etching and why HF dissolves oxide so readily but leaves the Si relatively untouched.

The preceding arguments give us a basic understanding of HF etching. As mentioned before in reality, however, the situation is much more complex, including HF, H_2F_2 , HF_2^- , F^- , H_3O^+ , OH^- , and sometimes NH_4F species together in the solution and in chemical equilibrium with one another. Additionally, there exist steric constraints at the surface and the effects of solvation on the reactions. A complete understanding of the detailed chemistry is not available at this time, but certain conjectures can be made with a reasonable degree of confidence. The calculations mentioned above were performed for molecules in free space so the energies need to be corrected to take into account solvation effects [60]. Furthermore, H_2O vapor is needed to initiate SiO_2 etching reactions with anhydrous HF [82,83], discussed in detail in Chapter 6. It can be argued that the main effect of placing the polar HF molecule into H_2O is to surround it, on the average, with H_2O molecules in the proper orientation to minimize the Coulomb energy. This, in turn, weakens the H—F bond, facilitating HF reactions that must break the HF bond. Therefore, the postulate is that solvation simply lowers the activation barriers that exist

for the gas-phase reactions described above. Reaction rate data are not available for the Si–Si back-bond attack, but HF dissolution of SiO₂ has an activation energy of approximately 0.35 eV [77], to be compared with the 0.55 eV calculated for the gas-phase reaction [60]. The heat of solvation to place an HF molecule into solution is ~ 0.4 eV and is consistent with the observed 0.2 eV lowering of the energy barrier. This point of view also allows rationalization of the HF and HF₂[−] etching behavior observed by Judge [77]. HF₂[−] can be thought of as a more highly solvated form of HF with a weaker bond strength that explains the lower activation energy for SiO₂ dissolution (0.31 eV) and the increased rate of dissolution (factor of 4–5). The role of steric hindrance at the surface is also important. In general terms, confidence should be placed on the chemical trends discussed above while remaining skeptical of the exact activation energies, because modifications caused by steric constraints or solvation can be expected.

One last point needs to be clarified regarding HF solution chemistry. It is now clear that the OH[−] concentration has a drastic effect on the etching that occurs with HF solutions [84]. Experiments show that H₂O rinsing alone can remove dihydride species at steps [84,85] and leads to monohydride termination on Si (111) surfaces [85]. In this regard, researchers noticed early on that samples etched in HF remained hydrophobic even after boiling in H₂O for extended periods of time [86]. It is equally curious that CMP Si wafers (polished in slurries with pH ~ 13) are hydrophobic and are terminated with H [87,88]. These observations taken together suggest that Si surface reactions with OH[−] can also lead to hydrophobic H-terminated Si surfaces once the surface oxide is removed. This leads to the speculation that HF and OH[−] chemistry may similarly remove Si atoms bonded to electronegative elements by back-bond attack of the polarized Si–Si bond. It also is interesting to note that HF and OH[−] in solution may have similarities in their reaction pathways at the surface.

9.3.3 Structure and Morphology of Hydrogen-Terminated Surfaces

This section starts with the effects of aqueous HF (concentrated and dilute HF solutions) on the surface morphology of Si (100) and Si (111) and then the etching of Si (111) and Si (100) in BHF solutions is separately examined. BHF is composed of various mixtures of 50 wt% (weight%) HF in H₂O and 40 wt% NH₄F in H₂O. A common mixture used in the industry is 7:1 BHF, which has a pH = 4.5 and is composed of 7 volumes of NH₄F and 1 volume of HF. The main difference between aqueous HF and BHF is the solution pH, which is the object of the discussion in Section 9.3.3. Throughout the subsections, the nature of the starting Si/SiO₂ interfaces and the role of H₂O (e.g., rinsing) in the etching process also are considered because they are relevant to the final surface structure.

9.3.3.1 Si (100) and Si (111) Etched in Aqueous HF Solutions

The first information available on the morphology of H-terminated Si (100) and Si (111) was the study by Hahn and Henzler [26]. Their goal was to use low-energy electron diffraction (LEED) to investigate the Si/SiO₂ interface morphology as a function of oxidation conditions and polishing methods, examining the diffraction peaks in detail; i.e., both their position and their line shape in reciprocal space. As with most surface analysis techniques, LEED requires SiO₂ film removal before the measurement. This is achieved by dissolving the oxide layer in concentrated HF followed by rapid loading into the analysis chamber. In this manner, the densities of step-atoms were studied as a function of oxidation conditions. An example of how the analyses were performed is found in reference [26], where LEED spot profile analysis was used to study the atomic structure of CMP Si (111) wafers. The original data, shown in Fig. 9.3-13, demonstrates that CMP Si (111) etched in concentrated

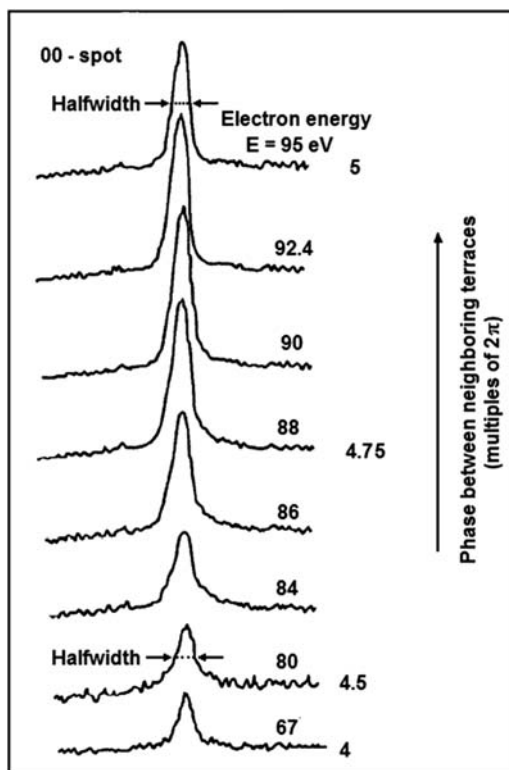


FIGURE 9.3-13 Low-energy electron diffraction spot profiles of a polished Si (111) wafer after HF treatment [89]. Used with permission from Materials Research Society. Reprinted from P.O. Hahn, *Mat. Res. Soc. Symp. Proc.* 54 (1986) 645, Pittsburg, PA.

HF is, on average, step-free over 100-Å distances [89]. In such an experiment, the electron wavelength is varied by varying the electron energy, leading to in- and out-of-phase scattering conditions from one atomic layer to another. Any broadening of the diffraction peaks would indicate a reduction of atomically flat terrace sizes. In the spectra shown in Fig. 9.3-13, the lack of observable broadening as a function of the electron wavelength indicates that the width is limited instrumentally and gives a lower limit for the average length of the terrace of 100 Å.

Next, Grundner and Schulz [69,70] used the vibrational frequency information from EELS on HF-treated Si (111) and Si (100) to investigate the nature of the H-termination. By monitoring the SiH₂ scissor mode of the dihydride species at 900 cm⁻¹, they concluded that Si (100) was dihydride terminated while Si (111) was monohydride terminated after HF etching. The EEL spectrum shown in Fig. 9.3-9 (top spectrum) for Si (100) is characterized by a strong loss peak at 900 cm⁻¹, which also is observed on clean Si (100) surfaces exposed to atomic H in UHV conditions. In all cases, the electron diffraction pattern measured by LEED is (1 × 1), i.e., there is no observable reconstruction. This observation led the authors to conclude that the surface is atomically well-ordered (unreconstructed) and, therefore, a uniform dihydride phase is obtained [30]. Similarly, the presence of a strong Si–H stretch loss at 2080 cm⁻¹ in Fig. 9.3-9 (bottom spectrum) and the observation of a good quality (1 × 1) LEED pattern led the researchers to conclude that the Si (111) surface was ideally monohydride terminated. The weak loss at 900 cm⁻¹ was attributed to dihydride at steps. These conclusions have since been shown to be oversimplified because (1 × 1) LEED patterns also are obtained when the top surface is disordered.

High-resolution IR reflection absorption spectroscopy is a powerful and complementary technique to EELS because it provides polarization information that is not available from EELS. Polarized spectra readily are measured using multiple internal reflection (MIR) IR spectroscopy and are particularly useful, for instance, to help elucidate the surface structure of H-terminated Si. Contrary to the conclusions drawn from EELS, the IR studies indicated that both Si (100) and Si (111) surfaces are atomically rough after HF etching treatments.

The roughness is evident from the complexity of the polarized IR absorption spectra shown in Fig. 9.3-14 for Si (111), for instance. An ideally terminated Si (111) surface would be characterized by a single monohydride stretch mode polarized normal to the surface (see Fig. 9.3-15) and this is not observed in Fig. 9.3-14. The spectra of Si (100) surfaces (not shown) also show that the HF-etched surfaces are much more complex than atomically flat, H-terminated surfaces. Although these surfaces are not ideally terminated, structural information can still be extracted through complete assignments of the observed bands. The mode assignments of the various hydride species can be performed by using isotopic (deuterium for example) substitution

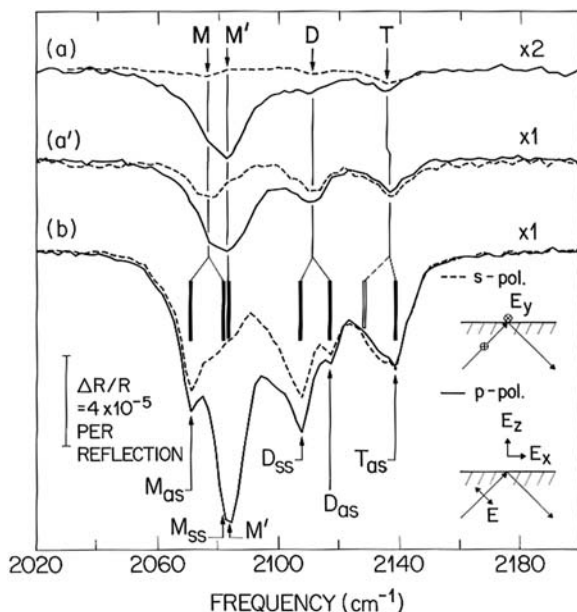


FIGURE 9.3-14 Polarized infrared absorption spectra of Si—H stretch vibrations after etching of a Si (111) sample in different isotopic concentrations of dilute HF/DF (deuterium fluoride) mixtures. The spectra (A) corresponds to 10% H and 90% D, (A') to 25% H and 75% D, and (B) to 100% H. The observed isolated frequencies are $M = 2077$, $D = 2111$, and $T = 2137 \text{ cm}^{-1}$. The thick vertical bars represent the calculated coupled mode splittings from the measured isolated frequencies (labeled M, D, and T) [72]. Reprinted with permission from V.A. Burrows, Y.J. Chabal, G.S. Higashi, K. Raghavachari, S.B. Christman, *Appl. Phys. Lett.* 53 (1988) 998. Copyright 1988, American Institute of Physics.

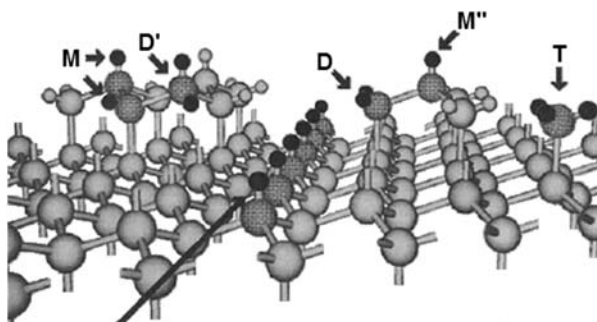


FIGURE 9.3-15 Schematic representation of possible Si (111) surface structures with H termination. The ideal monohydride and trihydride termination are possible for an atomically flat (111) plane. The “horizontal” dihydride (D) terminates the corner of a small structure where an isolated monohydride (M'') may exist. Both the “vertical” dihydride (D') and coupled monohydride (M) can terminate larger structures of the type shown here. These are all the possible structures that do not involve surface reconstruction [72]. Reprinted with permission from V.A. Burrows, Y.J. Chabal, G.S. Higashi, K. Raghavachari, S.B. Christman, *Appl. Phys. Lett.* 53 (1988) 998. Copyright 1988, American Institute of Physics.

experiments combined with force constant normal mode analyses on model compounds [73]. The Si—H stretch spectra in Fig. 9.3-1A and A' are simpler because deuterium is the major species, i.e., the H atoms are isolated from each other. The three main bands in the isolated spectra are assigned to the monohydrides (denoted as: coupled M or not coupled M'), dihydrides (O), and trihydrides (T). The corresponding spectra for the Si—D stretch (not shown) confirm these assignments. The solid vertical bars in Fig. 9.3-14 correspond to the theoretical predictions for the splittings associated with each hydride structure, starting from the measured value of the isolated frequency [72,73]. With these mode assignments, the structure of these H-terminated surfaces can be obtained. However, the assignment of trihydride (T) has been reconsidered as dihydride structures with strong steric interactions, which also have modes at that frequency.

For Si (111), the spectra in Fig. 9.3-14A are dominated by monohydride stretches, in agreement with the EELS data. In this case, part of the monohydride spectrum (M') is found to be polarized normal to the surface and can be associated with ideal termination (monohydride on a Si (111) terrace) [90,91]. Having identified the M' mode, the relative intensities of M, O, and T modes can be used to deduce the average surface structure, schematically represented in Fig. 9.3-15. In this figure, M terminates the side of a structure and T terminates the (111) terraces. There are two types of uncoupled monohydrides (M'), the ideal monohydride (M''), and two types of dihydrides (O and O'). M'' has not been measured separately, either because its frequency is too close to that of the M' and/or because its concentration is too low to be detectable. These data show that only a quarter of the surface is covered with M'. Step edge monohydrides (coupled monohydride, M) comprise another quarter of the surface. Together, M and M' make up approximately half of the surface hydrides leading to the strong monohydride EELS signature. The fact that EELS did not observe strong dihydride modes may appear contradictory if it were not for the fact that EELS suffers from problems with interpretation depending on whether the scattering mechanism is believed to be dipolar [92,93] or non-dipolar [94]. In contrast, the IR spectra are only sensitive to dipolar modes and therefore can give quantitative information on the dihydride stretch.

The main conclusion of these results is that Si etching in dilute HF leads to atomically rough surfaces as seen from the presence of mono-, di-, and possibly trihydrides on both Si (100) and Si (111) surfaces, indicating that the initial Si/SiO₂ interface is atomically rough. STM images of Si (111) [84] show structures of 10–20 Å diameter and 3 Å in height, accounting for about 50% of the surface, consistent with the IR data.

9.3.3.2 Si (111) Etched in Buffered HF Solutions

Raising the pH of the HF solution increases the etching rate of the Si surfaces [94]. For Si (111) surfaces, increasing the pH of the solution leads to faster

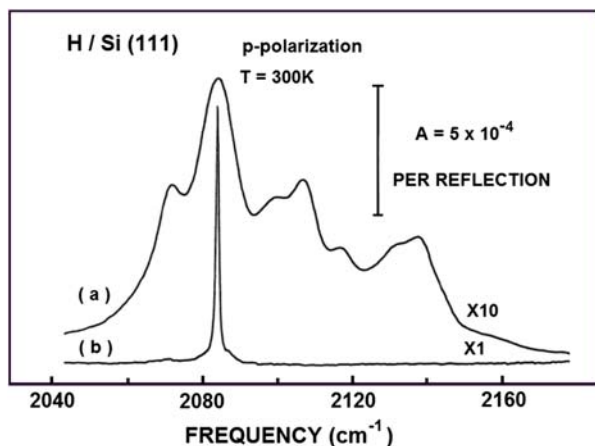


FIGURE 9.3-16 P-polarized infrared absorption spectra of Si (111) after (A) etching in dilute HF (pH = 2), and (B) a 40 wt% NH_4F solution (pH = 7.8) [95]. Reprinted from Y.J. Chabal, *Etching of silicon (111) and (100) surfaces in HF solutions: H-termination, atomic structure and overall morphology*, MRS Proc. 249 (1992) 349–360. Copyright (1992), reproduced with permission.

etching of defect structures, with the slowest etching occurring on the (111) planes, thus making it possible to flatten the (111) surface on an atomic scale [84]. For instance, Fig. 9.3-19 shows the difference between a Si (111) surface etched in dilute HF and buffered (pH $\sim$ 8) HF solutions (i.e., 40 wt% NH_4F). Whereas the dilute HF-etched surface is atomically rough, with all forms of hydrides, and whose spectrum is shown in Fig. 9.3-16A, the surface etched in a 40 wt% NH_4F solution is characterized by a single sharp absorption line at 2083.7 cm^{-1} , polarized perpendicular to the surface, shown in Fig. 9.3-16B [95]. The implication is that atomically flat surfaces are obtained with ideal monohydride termination. In Higashi's et al. [84] report, it is noted that the measured line width, $\Delta\nu$ of $\sim 0.9\text{ cm}^{-1}$, is the narrowest line ever measured for a chemisorbed atom or molecule on a surface at room temperature. Part of the width, however, can be because of thermal broadening. Therefore, a method was devised to introduce wafer samples into UHV, making it possible to cool them so that the inhomogeneous line width could be measured. The data confirm that most of the line width measured at room temperature is thermally induced because of anharmonic coupling of the Si–H stretch mode to surface Si phonons [91]. At present, the best samples are characterized by an extremely small (0.05 cm^{-1}) inhomogeneous broadening [90,96]. This result indicates a high degree of homogeneity and has motivated thorough characterization by many surface science techniques, such as XPS, EELS, IR, LEED, STM, and atomic force microscopy (AFM).

The LEED patterns obtained after careful introduction of the H-terminated silicon sample into UHV show a 1×1 pattern with resolution-limited integral-order spots and a background below the detection limit of conventional

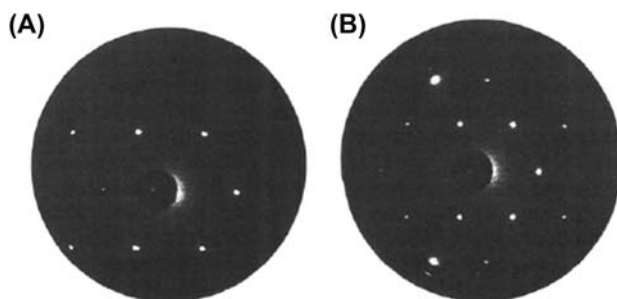


FIGURE 9.3-17 Photographs of low-energy electron diffraction patterns of Si (111) surfaces after NH_4F treatment: (A) 82 and (B) 125 eV [97]. Reprinted from G.S. Higashi, R.S. Becker, Y.J. Chabal, A.J. Becker, *Appl. Phys. Lett.* 58 (1991) 1656–1658. Copyright 1991, American Institute of Physics.

LEED systems as shown in Fig. 9.3-17. This unreconstructed and ideally H-terminated surface is often referred to as H/Si (111) (1×1).

STM images [97,98], such as the one shown in Fig. 9.3-18, have confirmed that the surface is nearly contamination free, atomically flat, and well-ordered with 1×1 ($3.84 \text{ \AA}$) periodicity. The electronic structure obtained from density of states (dI/dV) measurements displays no states in the gap, which is expected for H-terminated Si surfaces. Further support for the bulklike character and the ideal monohydride termination comes from electron-stimulated desorption experiments [99] showing the formation of π -bonded electron orbital chains, (2×1) reconstruction, after the H is desorbed. This surface also has been imaged with an AFM, confirming the 1×1 periodicity [100].

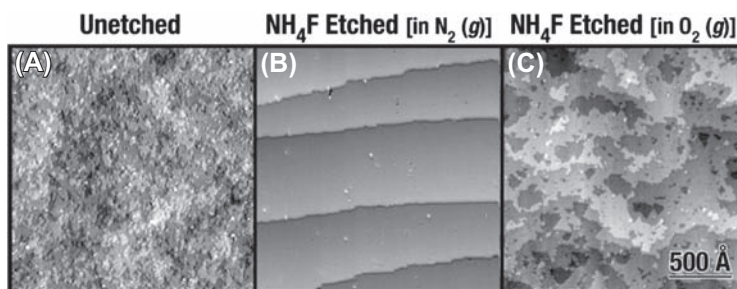


FIGURE 9.3-18 Scanning tunneling microscope images of Si (111) surfaces miscut by 0.3° toward the $11\bar{2}$ direction where (A) the surface is etched in buffered HF for 2 min under ambient conditions. This morphology reflects the roughness of the original Si/SiO₂ interface where (B) the surface is etched in quiescent 40 wt% NH_4F (aq) in a N_2 ambient and (C) the surface is etched in stirred 40 wt% NH_4F (aq) in an O_2 ambient. Dissolved O_2 leads to increased pit nucleation and a rougher etched surface compared with a N_2 environment [108]. Used with permission from M.A. Hines, *Int. Rev. Phys. Chem.* 20 (4) (2001) 645.

An alternate means of characterization is to measure the low-frequency surface vibrations (Si phonons) of this surface. Theoretical calculations of the vibrational manifold [101] predict strong dispersions for the modes of this ordered surface. Inelastic He atom scattering measurements, performed by Doak et al. [102], show phonon losses up to 30 meV (250 cm^{-1}). Two phonon branches (S8 and S'8) at 27.4 and 23.7 meV, respectively, are observed to disperse in addition, to the Raleigh wave between 0 and 16.3 meV, consistent with a well-ordered surface. The incoherent elastic scattering and broad inelastic backgrounds, arising from disorder, are almost two orders of magnitude lower for the H/Si (111) (1×1) than those measured for H-terminated Si (111) and Si (100) surfaces prepared using standard UHV techniques. This confirms the perfection of the surface compared with surfaces prepared in UHV.

EELS studies of the H/Si (111) (1×1) are similarly characterized by a very high specular beam intensity and a low background [75]. The dispersion of both the Si phonons and the H vibrations has been measured [94,103] and is in good agreement with the calculated curves [101].

High-resolution IR absorption spectra recorded at low temperatures ($<50\text{K}$) also have been used to characterize the extent of the perfect 1×1 domains [90,96]. Below 50K, the lifetime ($\sim 0.005\text{ cm}^{-1}$) [104] and thermal ($<0.001\text{ cm}^{-1}$) [91] broadenings are negligible compared with the measured line width (0.07 cm^{-1}), as shown in Fig. 9.3-19A. After deconvolution of the

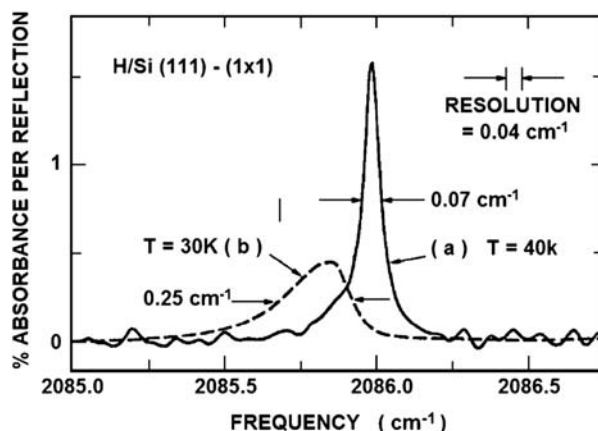


FIGURE 9.3-19 P-polarized infrared absorption spectra of H/Si (111) (1×1) prepared by: (A) thermal oxidation followed by etching in buffered HF (pH ~ 5) for 2 min and subsequent etching in 40 wt% NH_4F for 4 min, and (B) chemical oxidation followed by etching in a 40 wt% NH_4F solution for 6 min. Both samples were thoroughly rinsed in DIW after the last etching step [96]. Reprinted from P. Jakob, P. Dumas, Y.J. Chabal, *Appl. Phys. Lett.* 59 (1991) 2968. Copyright 1991, American Institute of Physics.

resolution function (0.04 cm^{-1}), the natural line width (0.05 cm^{-1}) and the line shape are obtained and can be related directly to surface inhomogeneity.

The surface structure also was studied using detailed line shape analysis of the IR absorption bands associated with Si–H vibrations by Jakob et al. [90]. In particular, line shapes were analyzed with regards to both a distribution of point defects leading to symmetrical broadening (such as Lorentzian or Gaussian) and the presence of finite-size domains (flat terraces) leading to asymmetric broadening. When both effects are taken into account, the spectra of Fig. 9.3-19A can be fitted with $N = 2 \times 10^4$ Si–H units with a 30% distribution in domain (terrace) size [90]. STM images show that the average linear terrace size is 500 Å on flat samples, which is in excellent agreement with the IR absorption line shape analysis (~ 600 Å) [105].

STM images of NH_4F -etched Si (111) surfaces show a wide variation in point defect densities. The origin of these variations was first elucidated by Wade and Chidsey [106] who showed that a low concentration ($\sim$ ppm, parts per million) of dissolved O_2 in the etchant would lead to etch pit nucleation. Once nucleated, these pits grow and lead to rough etched surfaces characterized by equilateral triangular pit shapes [107,108]. In a pure O_2 ambient, dissolved O_2 leads to pronounced roughening as shown in Fig. 9.3-18.

The sensitivity of the IR absorption technique makes it possible to investigate in detail the atomic roughness resulting from various preparations of the surface. In particular, the nature of the oxide before the HF etching has been found to be important. Fig. 9.3-19 shows high-resolution IR spectra associated with H/Si (111) (1×1) surfaces prepared in two different ways: (1) by stripping the thick thermal oxide in BHF ($\text{pH} = 5$) for 2 min and then directly dipping the H-passivated sample into a 40 wt% NH_4F solution for 4 min and (2) by chemically reoxidizing (SC-2) and then stripping and re-etching in a 40 wt% NH_4F solution for 6 min. The first absorption line was discussed previously in this section. The second spectrum peaks at lower frequency, indicates a smaller average terrace (domain) size. It is also broader, indicating a larger distribution of domain sizes. Therefore, thermally grown oxides result in a smoother interface (i.e., larger domains of monohydrides) than chemically grown oxides. Here, the difference in surface morphology results from variations in the etching rates of HF versus NH_4F .

In fact, the variation of etching rates is a well-known phenomenon. This phenomenon has been investigated further by treatment in concentrated HF where the etching rate of silicon is minimal. Previous researchers have found that, when a thick thermal oxide (~ 1000 Å) is grown on Si (111) at a temperature of $\sim 1050^\circ\text{C}$, followed by postannealing in an inert gas, a very smooth Si/SiO₂ interface is formed [3,26,36,109]. Dissolution of this oxide in concentrated HF produces an H-terminated Si (111) surface, characterized by a complex spectrum with multiple peaks, which indicates atomic roughness, as shown in Fig. 9.3-20A. This roughness disappears upon rinsing as shown in Fig. 9.3-20B as evidenced by the dominance of the monohydride peak, at

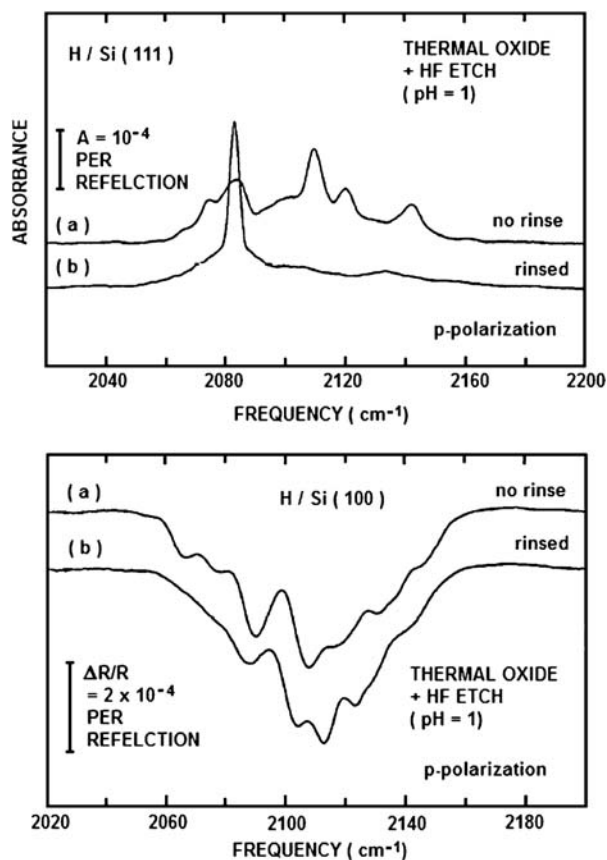


FIGURE 9.3-20 Top: P-polarized infrared (IR) absorption spectra of thermally oxidized Si (111) after (A) etching in concentrated HF, and (B) subsequent rinsing in DIW. Bottom: P-polarized IR absorption spectra of thermally oxidized Si (100) after (C) etching in a concentrated HF solution and (D) subsequent rinsing in DIW. *Unpublished, used with permission from Y. J. Chabal.*

$\sim 2080 \text{ cm}^{-1}$. In contrast, the complex spectrum associated with atomically rough Si (111) surfaces produced by HF etching of a chemical oxide is not removed upon simple rinsing, indicating that the atomic roughness originally present at the Si/SiO₂ interface is not removed.

These observations confirm that the thermally grown oxide has an atomically smoother interface than the chemically grown oxide and suggests that H₂O rinsing alone can preferentially remove small surface defects. Identical experiments also were performed on Si (100) surfaces and are shown in Fig. 9.3-20, bottom spectra. Unlike the sharp spectra observed for Si (111) surfaces, multimode spectra are obtained for Si (100), which suggests that the surfaces are atomically rough. Furthermore, certain spectral changes occur

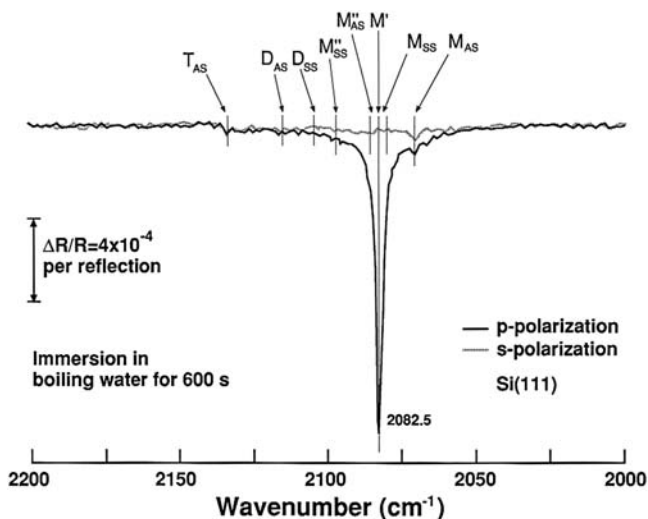


FIGURE 9.3-21 Polarized infrared absorption spectra of a chemically oxidized Si (111) sample after etching in a 1.5 wt% HF solution and subsequent boiling in DIW at 100°C for 10 min. The resolution is 0.5 cm^{-1} [85]. Reprinted with permission from S. Watanabe, N. Nakayama, T. Ito, *Appl. Phys. Lett.* 59 (1991) 1460. Copyright 1991, American Institute of Physics.

after rinsing. The monohydride modes decrease in strength while those of the dihydride modes increase. Note that this is opposite to the behavior observed on Si (111) surfaces.

Preferential etching by H_2O , resulting in flat H-terminated Si (111) surfaces, has been demonstrated by Watanabe et al. [85]. These authors studied the effects of H_2O rinsing as a function of H_2O temperature, finding that hot H_2O (100°C) increases the rate of removal of (111) surface defects (pits, steps, and kinks) while maintaining the H-termination. The resulting spectrum, shown in Fig. 9.3-21, features a single-mode, polarized perpendicular to the surface, which dominates the spectrum. However, using the spectral analysis developed by Jakob et al. [90], the average (111) terrace size is found to be 20 Å, a factor of 15 smaller than the sample shown in Fig. 9.3-19A. Nonetheless, it is surprising that boiling H_2O rinsing does not lead to the growth of a surface oxide. A possible mechanism is discussed in Section 9.3.4.

9.3.3.3 Si (100) Etched in Buffered HF Solutions

The discovery of atomically flat H-terminated Si (111) surfaces prepared via wet chemical etching led to the desire to prepare analogous flat Si (100) surfaces because these are more technologically relevant to the microelectronics industry. As for the Si (111) surfaces, it was found that raising the pH of the HF solution increases the etching rate of the H-terminated Si (100) surfaces [94]; however, for the Si (100) surfaces, a more complex surface

morphology is observed, which critically depends on the pH of the HF solution and the reaction conditions. In this section, the results from the 1990s using standard BHF solutions are presented first, in which the interpretation of the spectra was associated with a combination of mono-, di-, and trihydride terminations [73,94]. Next examined will be how, by controlling the oxygen content of the fluoride solution and the by-products of the etching process (i.e., H_2 gas evolution), the morphology of H-terminated Si (100) surfaces can be tuned from microfaceted to near-atomically flat [54,110,111].

The effect of solution pH is observed in Figs. 9.3-22 and 9.3-23, which shows that the morphology of chemically prepared Si (100) surfaces changes as the pH of the etching solution varies from 2 to 8 [94]. For a pH = 2 solution (Fig. 9.3-22A), the IR absorption spectra are dominated by dihydrides between 2100 and 2120 cm^{-1} . Whereas, in BHF (pH $\sim$ 5), the spectrum sharpens and is dominated by coupled monohydrides appearing as modes at 2073 and 2087 cm^{-1} (Fig. 9.3-22B). For higher pH values, the etching proceeds quickly, as evidenced by the gas bubbles that form at the sample surface. After etching in an NH_4F solution (pH = 8), the dihydride contribution is again dominant (Fig. 9.3-23A).

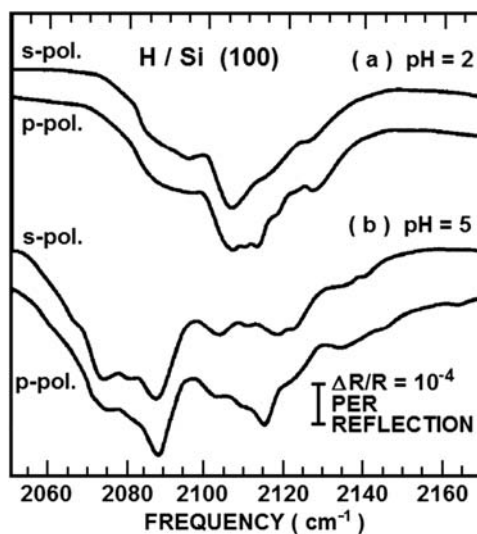


FIGURE 9.3-22 Polarized infrared absorption spectra of Si (100) surfaces etched in: (A) dilute HF (1 wt%, pH = 2) and (B) buffered HF (pH = 5). The chemically oxidized surface is used as a reference and the spectral resolution is 1 cm^{-1} [94]. Adapted from P. Dumas, Y.J. Chabal, P. Jakob, *Morphology of hydrogen-terminated Si (111) and Si (100) surfaces upon etching in HF and buffered-HF solutions*, *Surf. Sci.* 269–270 (1992) 867. Copyright (1992), with permission from Elsevier.

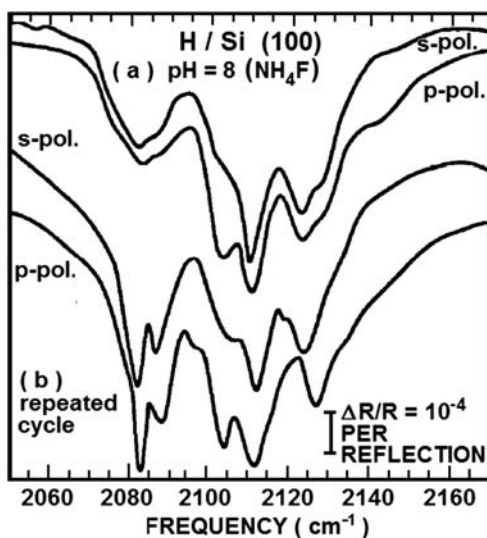


FIGURE 9.3-23 Polarized infrared absorption spectra of (A) a Si (100) surface etched in a 40 wt % NH_4F solution ($\text{pH} = 8$) once and (B) after repeated cycles of etching in a 40 wt% NH_4F solution, chemical reoxidation in $\text{H}_2\text{O}_2\text{:HCl:H}_2\text{O}$ (1:1:4) at 80°C , and a final etching in NH_4F [94]. Adapted from P. Dumas, Y.J. Chabal, P. Jakob, *Morphology of hydrogen-terminated Si (111) and Si (100) surfaces upon etching in HF and buffered-HF solutions*, *Surf. Sci.* 269–270 (1992) 867. Copyright (1992), with permission from Elsevier.

The dihydride-dominated spectra shown in Fig. 9.3-22A ($\text{pH} = 2$) are the result of treatment in dilute HF, which represent a surface that is rough on an atomic scale, as previously discussed. The dominance of the monohydride modes in Fig. 9.3-22B suggests the formation of microfacets on the Si (100) surface [84,94]. At high pH ($\text{pH} = 8$, Fig. 9.3-23A), the spectra revert back to being dominated by the dihydride stretch, however, the polarization of this mode is quite different from the $\text{pH} = 2$ spectra. In this case, the symmetric stretch (2105 cm^{-1}) is polarized normal to the surface and the anti-symmetric stretch (2112 cm^{-1}) is polarized parallel to the surface.

After several etching cycles, Fig. 9.3-23B shows that two sharper modes are resolved at 2084 and 2088 cm^{-1} . The first is assigned to the Si–H stretch of the ideal Si (111) monohydride, confirming that (111) facets develop in high pH solutions. The second is probably associated with the symmetric stretch of coupled monohydrides; the asymmetric stretch mode is more highly screened and therefore not observed. Both types of monohydrides have symmetric stretches pointing away from the normal of the macroscopic surface plane and are therefore unscreened. In contrast, the dihydride modes in Fig. 9.3-23 are characteristic of dihydrides with their axis pointing along the surface normal. The simplest atomic arrangement consistent with these observations is a distribution of tentlike structures with a row of dihydrides at the rooftop, (111)

facets terminated with ideal monohydride on the sides, and coupled monohydride at the periphery of the facets. Because the facets are small, the concentration of coupled monohydrides is as high as that of ideal monohydrides. In fact, the production of this type of structure by aqueous etching was confirmed by Skibinski and Hines via STM imaging, molecular modeling, and IR spectroscopy [112].

The increased surface roughness upon etching that is seen through spectroscopic methods is correlated with microscopic surface analysis. AFM carried out before and directly observed after BHF etching highlights the increase in surface roughness [6]. As shown in Fig. 9.3-24, a control wafer is relatively smooth with $\sim 2 \text{ \AA}$ RMS roughness, whereas a wafer treated in BHF is characterized by $\sim 5 \text{ \AA}$ RMS roughness. In attempting to smooth Si (100)

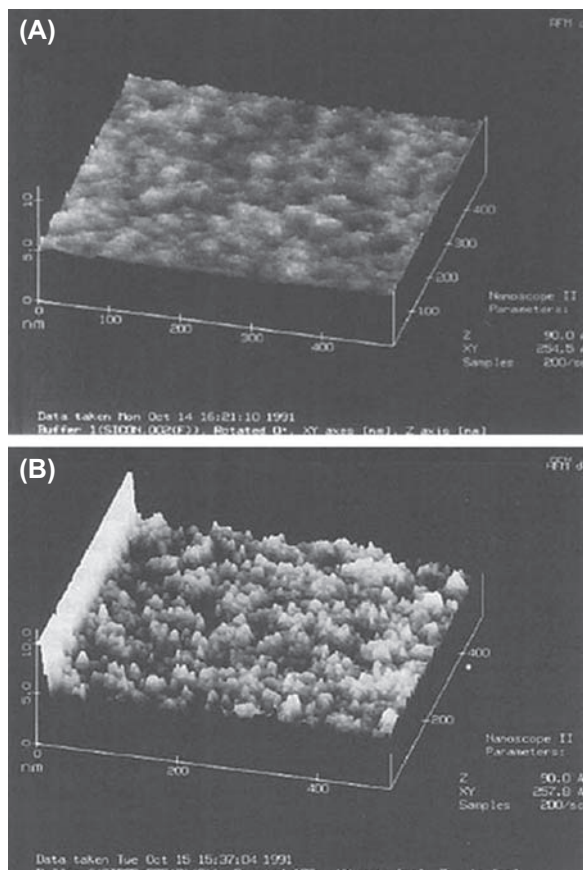


FIGURE 9.3-24 Atomic force microscope images of (A) a chemical mechanical polishing Si (100) control wafer ($\sim 2 \text{ \AA}$ RMS) and (B) an Si (100) wafer etched in 7:1 buffered HF solution for 10 min ($\sim 5 \text{ \AA}$ RMS). Used with permission from J. Sapjeta, unpublished.

surfaces, thermal oxidation, which is known to result in high-quality Si/SiO₂ interfaces, may be used [3,26,36,109]. In such an experiment, a 1000-Å thick dry O₂ oxide is grown at 1000°C and postannealed in Ar at the same temperature for 30 min. The oxide is then removed with concentrated HF. The resulting complex spectra suggest that the surfaces are still atomically rough.

These initial results suggested that BHF may not be as good as ammonium fluoride to prepare atomically flat (100) surfaces, because (111) facets develop upon etching. In fact, BHF is only used in research activity. The formation of these surface facets was thought to arise because of steric hindrance [113]. In fact, first principle calculations found that a perfectly flat Si (100) passivated with rows of dihydrides would be so highly strained that the dihydrides would need to cant significantly, ~ 15 degrees, to partially relax the strain as seen in Fig. 9.3-25 [114,115].

However, recent work has shown that steric hindrance does not drive surface roughening, but rather it is caused by an unintentional side product of the etching mechanism. Many etches produce H₂ gas bubbles, which are often pinned on the surface and result in morphological changes [110]. If the gas bubbles are allowed to remain and coalesce on the surface, circular features (pits and pillars) and faceted micropits are present on the surface after etching [110]. However, if the bubble formation and coalescence is prevented by either sonication or pulling across etchant-air interface, near-atomically flat, or “half-flat,” H-terminated Si (100) surfaces are obtained as determined via STM, shown in Fig. 9.3-26 [110,111]. From the STM images, the surface is seen to be dominated by long rows that are 0.9 Å tall and spaced by ~ 7.7 Å, which corresponds to twice the distance between two adjacent Si atoms on the bulk surface. Additionally, elliptical protrusions, separated by ~ 3.8 Å, that are perpendicular to the rows are observed. These features rotate by 90 degrees on neighboring terraces. This morphology results in a very smooth surface with an RMS roughness of 0.57 Å.

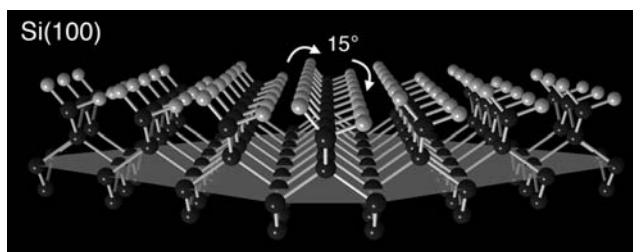


FIGURE 9.3-25 Molecular models of atomically flat H-terminated Si (100). Calculations found that to relieve strain, the dihydrides would need to cant by 15 degrees [54]. Reprinted with permission from E.S. Skibinski, M.A. Hines, *Finding needles in haystacks: scanning tunneling microscopy reveals the complex reactivity of Si (100) Surfaces*, *Acc. Chem. Res.* 48 (2015) 2159–2166. Copyright (2015) American Chemical Society.

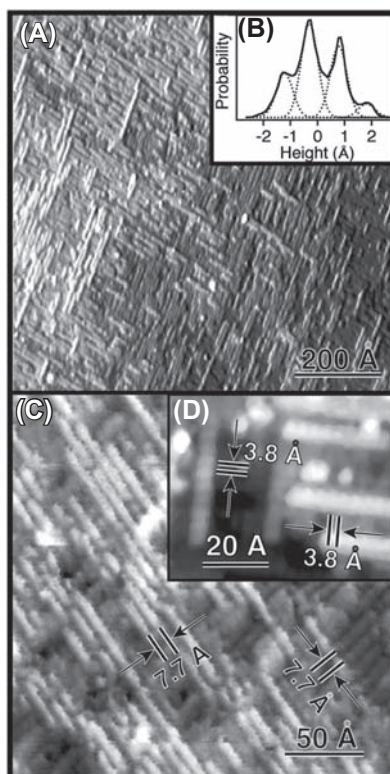


FIGURE 9.3-26 Scanning tunneling microscope images of a Si (100) surface etched in NH_4F [94]. The surface has a near-atomically flat morphology with an RMS surface roughness of $0.57 \text{ \AA}$. Height distribution of (A) is shown in the inset [111]. Reprinted with permission from I.T. Clark, B.S. Aldinger, A. Gupta, M.A. Hines, *Aqueous etching produces Si (100) surfaces of near-atomic flatness: strain minimization does not predict surface morphology*, *J. Phys. Chem. C* 114 (2010) 423–428. Copyright (2009) American Chemical Society.

To understand the chemical structure of the surface, polarized IR absorption spectroscopy was performed. Surprisingly, the IR spectra of the “half-flat” surfaces (not shown) actually appear quite similar to rough surfaces measured earlier [94] with multiple distinct modes, which is in contrast with the STM images that show a near-atomically flat surface [111]. To understand how these spectra correspond to the observed “half-flat” surface, deconvolution of the s- and p-polarized spectra into their Cartesian components was carried out [116]. However, it was found that insight into the origin of the Si–H modes is obscured by the four-fold symmetry of the surface, which makes it impossible to distinguish components of the dipole moments in the x-direction (μ_x) and y-direction (μ_y). To overcome this obstacle, vicinal Si (100) surfaces, for which the symmetry is broken, are utilized. The deconvoluted spectra for the

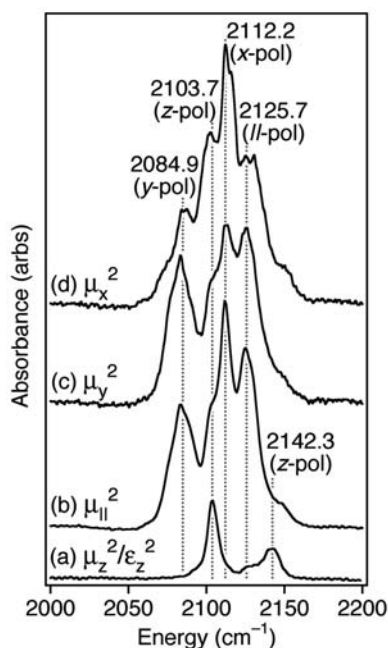


FIGURE 9.3-27 The Cartesian components of the squared transition dipole moment in the Si–H stretch region extracted from infrared spectra of NH_4F -etched Si (100) surfaces: (A and B) flat and (C and D) miscut by 5° toward [011] crystal orientation [111]. Reprinted with permission from I.T. Clark, B.S. Aldinger, A. Gupta, M.A. Hines, *Aqueous etching produces Si (100) surfaces of near-atomic flatness: strain minimization does not predict surface morphology*, *J. Phys. Chem. C* 114 (2010) 423–428. Copyright (2009) American Chemical Society.

flat and vicinal surfaces are shown in Fig. 9.3-27 where five distinct modes are visible [111]. The modes represent vibrations that are either oriented perpendicular or in-plane polarized. The mode at 2084.9 cm^{-1} is in-plane polarized, which goes against the theory that the surface contains Si (111) microfacets because for this vibration to represent a monohydride, it would be oriented perpendicular to the surface.

With the knowledge gained from the STM images and IR data, a molecular model of the surface was developed, shown in Fig. 9.3-28 [54,111]. The surface consists of single atom-wide rows of unstrained dihydrides characterized by modes at 2103.7 cm^{-1} (x-polarized) and 2112.2 cm^{-1} (z-polarized) for the asymmetric and symmetric vibrations, respectively. The single-atom deep trenches are terminated by strained dihydrides that are rotated 90° with respect to the unstrained dihydrides. These strained dihydrides are canted because of steric interactions as in Fig. 9.3-25A and the strained dihydride modes at 2084.9 cm^{-1} (y-polarized) and 2142.3 cm^{-1} (z-polarized) are associated with the lower and upper Si–H bonds, respectively. The mode at 2925.7 cm^{-1} ,

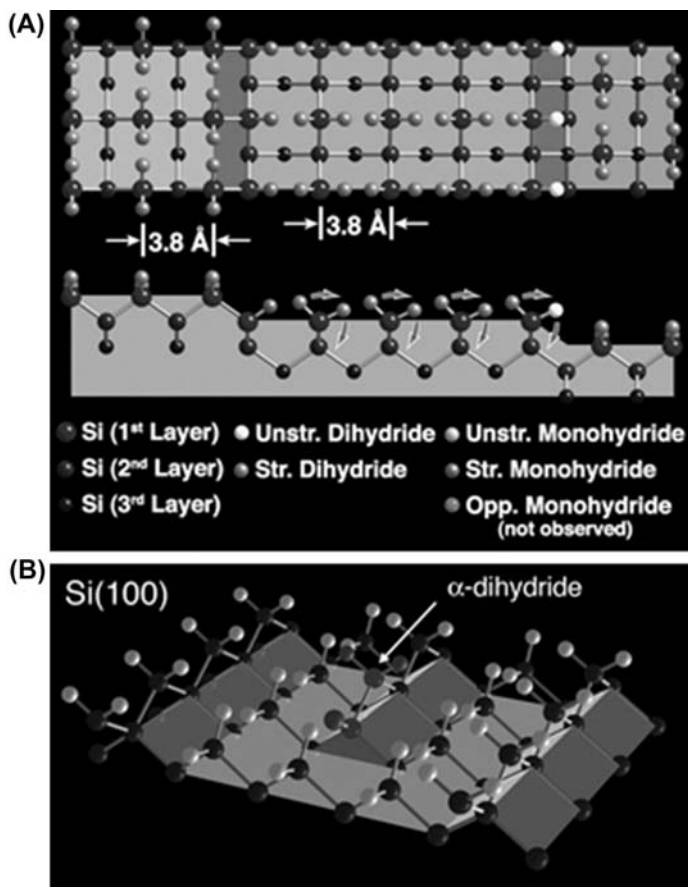


FIGURE 9.3-28 A schematic of the (A) top view and (B) cross-sectional view of “half-flat” surface structures showing atomic configuration resulting from an NH_4F etch. A schematic of the (C) different orientations of the ideal Si (100) with $-\text{Si}-\text{H}_2$ bonding produced by etching in NH_4F . These figures are found in references [45,111]. (A) Reprinted with permission from I.T. Clark, B.S. Aldinger; A. Gupta, M.A. Hines, *Aqueous etching produces Si (100) surfaces of near-atomic flatness: strain minimization does not predict surface morphology*, *J. Phys. Chem. C* 114 (2010) 423–428. Copyright (2009) American Chemical Society. (B) Reprinted with permission from E.S. Skibinski, M.A. Hines, *Finding needles in haystacks: scanning tunneling microscopy reveals the complex reactivity of Si (100) Surfaces*, *Acc. Chem. Res.* 48 (2015) 2159–2166. Copyright (2015) American Chemical Society.

polarized parallel to the surface, is assigned to strained monohydrides that form as a result of surface imperfections.

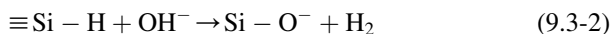
The knowledge gained from understanding the surface morphology provides insight into the etching rates for different species. A mechanism for anisotropic etching of Si (100) surfaces is discussed in [Section 9.3.4.2](#). From

earlier work, a clear dependence on the pH of the etch solution is observed [94]. Knowing that “half-flat” H-terminated Si (100) surfaces could be prepared, the effect of etchant pH was investigated to understand its effect on etching rates and morphology [117]. This knowledge is especially important because the etchant used in the IC industry is usually mildly acidic (pH = 5) rather than basic (40% NH_4F has a pH = 7.8). Aldinger et al. [117] varied the pH from 5 to 11 to investigate the changes in surface morphology via STM and IR spectroscopy. They found that at $\text{pH} \geq 7.8$, near-atomically flat surfaces are produced; however, from $7.5 \geq \text{pH} \geq 5$ significant changes are observed, such as microfacets. In the IR spectra of these low pH etches, three new Si–H modes are observed that correspond Si–H bound to Si (110) microfacets. The formation of these microfacets is confirmed by the observation of pyramidal hillocks with STM. Hillock formation is believed to arise from by a competing etching mechanism, similar to that on Si (111) surfaces, that dominates at low pH environments.

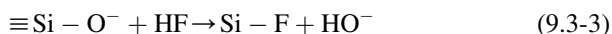
In summary, Si (100) surfaces are microscopically rough when treated in either dilute or concentrated HF. However, by carefully controlling the reaction conditions (i.e., pH) and avoiding bubbling at the surface, atomically flat or “half-flat” surfaces can be produced. Additionally, high pH is critical to prevent surface roughing.

9.3.4 Mechanism of Preferential Etching

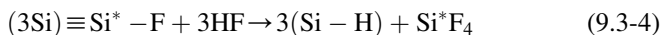
A solution of concentrated HF dissolves SiO_2 efficiently and passivates the Si surface with H. Once H-passivation is achieved, etching stops. As a result, the morphology of the original Si/ SiO_2 interface is preserved. The data presented in the previous section indicate that diluting concentrated HF with H_2O or buffering with NH_4F induces a slow etching reaction yielding H-passivated Si surfaces. The overall etching rate increases with the pH of the solution, as evidenced by the increased formation of small H_2 bubbles as the pH is raised. The bubbles are probably formed during oxidation of the surface by OH^- , according to the following reaction:



Once oxidized, the surface is subject to HF attack through HF insertion into the Si–O bond, according to the schematic reaction:

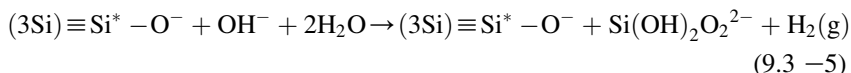


with subsequent removal of the surface Si atom (now labeled Si^* to distinguish it from the underlying bulk Si atoms) and passivation of the second-layer Si atoms by H, according to the mechanism proposed in Section 9.3.2:



In the above processes, the last two steps are fast compared with the initial oxidation of the H-passivated surface. As a result, the surface is always H-terminated.

The role of OH^- is a key ingredient in the attacking and etching of H-terminated Si surfaces [60]. It also is important to note that Si can be etched without HF. For instance, silicon can be etched with alkaline solutions, such as KOH or NaOH [118,119] and even with H_2O [85]. These observations indicate that, once oxidized, the Si surface can be attacked by OH^- groups. A plausible reaction pathway involves the Si back-bond attack by OH^- :



Confirmation and quantification of the above reaction steps should be possible using first principles (i.e., *ab initio*) cluster calculations, as was done to understand the H-passivation of Si [60]. In addition, the influence of surface charges on the anisotropic etching of Si needs to be understood [120].

The key point in considering preferential etching is that oxidation of H-terminated Si surfaces is extremely slow and is the rate-determining step. It takes many collisions between OH^- ions and surface Si–H groups to effect a reaction because the reaction barrier is large. When this is the case, relatively minor factors may strongly influence the reaction probabilities. For instance, if some surface structures are strained, they may be more easily attacked because the reaction barrier is lowered. Alternatively, if a surface structure is more accessible to OH^- ions in solution, it may be attacked faster because of an increased reaction probability.

9.3.4.1 Mechanism for Si (111) Surfaces

To understand and quantify the etching rates of various surface structures, the chemical etching of stepped Si (111) surfaces was studied by Jakob et al. [90] by utilizing IR absorption spectroscopy and STM images to characterize the surface structures after each chemical treatment. The results are summarized in the schematic drawing of Fig. 9.3-29 [121]. At low pH (pH = 1–3), the HF solutions strip the oxide from the original Si/SiO₂ interface but do not substantially etch the Si (111) surface, which usually displays a fair degree of atomic roughness. As the pH is increased (pH > 6.6), smoother surfaces result. The original model of these results proposed by Jakob and Chabal attributed the variations in surface roughness to the effects of solution pH [113]. However, more recent work has shown that the changes in surface morphology are a result of more subtle effects [122].

As the pH of the fluoride solution is lowered from pH = 7.8 (i.e., pure NH_4F), the Si etching rate drops linearly with $[\text{OH}^-]$ as predicted by Eq. (9.3-2). If there is dissolved O_2 in the solution, a second reaction becomes important: the oxidation of H–Si by O_2 . This oxidation occurs randomly

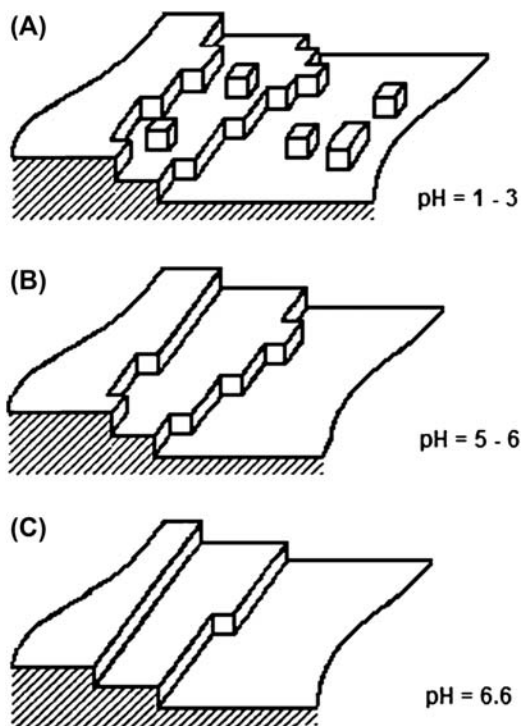


FIGURE 9.3-29 Schematic picture of the changes of the Si surface morphology as the etching rate is increased by increasing the pH of the etching solution: (A) $\text{pH} < 3$, (B) $\text{pH} = 5-6$, and (C) $\text{pH} = 6.6$. A total etch time of 3 min, including the removal of approximately $10 \text{ \AA}$ SiO_2 , is assumed. A pH higher than 6.6 leads to step bunching and therefore to the formation of multiple steps and facets (not shown here) [121]. Reprinted from P. Jakob, Y.J. Chabal, K. Raghavachari, R.S. Becker, A.J. Becker, *Kinetic model of the chemical etching of Si (111) surfaces by BHF solutions*, *Surf. Sci.* 275 (3) (1992) 407. Copyright (1992), with permission from Elsevier.

across the surface. Once a site becomes oxidized, it is then rapidly etched by the HF. The issue with O_2 oxidation was first recognized by Wade and Chidsey in 1997 [106] and its impact on surface morphology was later shown via STM images [122]. With these two competing factors, the observed surface morphology at different pH results from the interplay between the kinetics of the O_2 and OH^- in solution. The rough surfaces observed at low pH are a result of isotropic reactivity of O_2 which dominates in this regime. On the other hand, as the pH increases, the $[\text{OH}^-]$ increases and therefore, the anisotropic behavior of OH^- dominates [52,122].

The above observations indicate a “step flow” etching mechanism, whereby etching of kinks is fastest, leading to unzipping of the step atoms [98]. As the pH is increased beyond 7.0, the etching of stepped surfaces

increases drastically as evidenced by the formation of bubbles. The surface then roughens, partly because of step bunching [118] and partly because of the more inhomogeneous conditions at the surface (e.g., bubbles, fluctuation in the concentrations of various chemical species). The result is the formation of large, three-dimensional roughness as evident in STM images [105].

The site-specificity of NH_4F etching reactions is quantified using a combination of IR spectroscopy, STM, and kinetic Monte Carlo simulations [52,108]. As expected, this analysis confirms that NH_4F is a highly anisotropic etchant, which attacks all defect sites (e.g., step sites, kink sites) at least 5000 times more rapidly than terrace sites. The most reactive site, the kink site, is at least 1×10^7 times more reactive than the least reactive site, the terrace site. The correlation between the structure and reactivity of the various sites provide insight into the reaction mechanism.

The observed trends in reactivity are consistent with a reaction that is rate-limited by a pentavalent transition state, as first proposed by Hines et al. [123]. For example, the rigid, tetrahedral geometry of the terrace and monohydride step sites leads to a relatively low etching rate as the three Si–Si bonds are strong and hard to distort. In contrast, the highly strained kink site, which is held in a near-pentavalent geometry, reacts much more rapidly. In short, the reactivity of the surface appears to be dominated by structural considerations.

The chemistry on these surfaces is obviously complex. The simple mechanisms described above are meant only to give a framework in which to address the problem. The etching reactions are also surprisingly sensitive to inhomogeneity of the etchant solution [124] and ppm levels of contamination [106].

9.3.4.2 Mechanism of Si (100) Etching

Unlike Si (111) surfaces in which step flow models for the etching mechanism and insight into reactivity rates have existed for many years [98,108], the observed “half-flat” H-terminated Si (100) surface was a completely unknown structure. In fact, before its experimental observation, this structure had not been predicted to form [111,114,115]. From the resulting H-terminated surface morphology shown schematically in Fig. 9.3-28A, some insight into the reactivity of the different sites can be inferred. The significant presence of strained dihydrides implies these sites must be fairly unreactive or else they would have been etched away to form rows of strained monohydrides. Additionally, the formation of long rows of unstrained dihydrides suggests that the coupled strained monohydride–dihydride terminal unit is highly reactive, especially when compared with the rows. Lastly, the presence of missing rows reveals that strain plays a significant role in determining the surface morphology [111].

To understand how this surface could be prepared and to gain insight into most reactive site, kinetic Monte Carlo simulations were carried out on both

flat and vicinal surfaces in which the etching rates of the different surface sites were varied [54,125]. Using this, tens of thousands of potential surfaces were created, and the best fit to the spectroscopic and morphological data was determined. The relative reactivities of all of the sites on the Si (100) surface were estimated from the best fit and used to determine the etching mechanism that produces these near-atomically flat surfaces.

The most reactive site on the Si (100) surface is found to be an unstrained dihydride coupled to a neighboring monohydride, a so-called “ α -dihydride,” which is commonly found on the end of rows of unstrained dihydrides as shown in Fig. 9.3-28B. In fact, this site is 1000 times more reactive than unstrained dihydrides in the middle of the row [125]. The high reactivity of this α -dihydride is because of its bonding structure and location. To etch a Si atom from the surface, two back-bonds need to be broken in sequential reactions. The key point is that the first back-bond must be broken *and passivated* before the second bond is broken. The α -dihydride is unique in that cleavage of the bond between the α -dihydride and the adjacent monohydride allows the monohydride to pivot and release strain. This ability to relax makes the α -dihydride the most reactive surface site. Additionally, the etching of the row-end α -dihydride is self-propagating as the removal of each row-end site results in the formation of a new α -dihydride on the neighboring dihydride. This chain reaction accounts for the formation of single-atom-wide rows on the surface.

Although the reactivity of the α -dihydride explains the formation of single rows, it does not explain the alternating row structure, which was observed both experimentally and theoretically. The appearance of this surface structure again results from steric hindrance. In this case, the etching of a strained dihydride distorts the geometry of the neighboring dihydrides and impacts the interadsorbate strain. In fact, the removal of the most reactive site results in the release of maximum energy by the neighboring dihydrides upon reduction of interadsorbate strain as a result of distortion caused by etching. The effect of the interadsorbate strain leads to the observed surface morphology [54,125].

The resulting morphology of these surfaces is complex; however, with a combination of spectroscopic, microscopic, and theoretical techniques insight into the mechanism and site specific reactivity can be obtained.

9.4 CONTAMINATION AND STABILITY OF HYDROGEN-TERMINATED SURFACES

9.4.1 Oxidation of Hydrogen-Terminated Silicon Surfaces

Oxidation of H-terminated Si surfaces is particularly important to understand because epitaxially grown Si and SiGe source/drains, introduced into manufacturing at approximately the 90-nm node, require H-passivation that not only covers 100% of the surface but survives long enough to meet

manufacturing queue time requirements. Trace contamination caused by oxidation will be discussed in detail in the following sections, with much of the material taken from a review article [126].

The barriers to oxidation of H-passivated surfaces are high. In pure environments of O_2 , N_2 , or H_2O , the surface is characterized by a low reactivity that requires high exposures and relatively high temperatures to achieve oxidation. To study the oxidation mechanism, it is important to determine the impact of (1) chemical and structural surface defects and (2) impurities in the reactant, because such imperfections may lead to dramatically higher reactivity. These issues have been addressed by using (1) surfaces with different defect types and densities and (2) high-purity reactants.

In Section 9.4.2 results are summarized on the interaction of H-terminated Si with liquid H_2O , because a H_2O rinse is an inherent part of H-terminated Si fabrication. Also, comparison of wet and dry reactivity of H-terminated Si is discussed because this knowledge will aid in a full understanding of the mechanistic of oxidation. Section 9.4.3 highlights high-purity gas phase studies involving O_2 and H_2O , which are the most relevant oxidants for high-k gate oxide processes. The kinetics and barriers extracted from these studies can then be invoked to model oxidation and ALD growth. Section 9.4.4 briefly reviews ambient oxidation studies. Such results are not only important to assess the impact of H-terminated Si contact with clean room environments but on a fundamental level also help to understand the mechanistic impact of impurities such as H_2O in O_2 vapor. Lastly, Section 9.4.5 briefly examines the effect of other surface contaminants such as residual F, O, or hydrocarbons on the quality of the H-termination.

9.4.2 Aqueous Chemistry of Hydrogen-Terminated Silicon and the Role of Dissolved Oxygen

The aqueous chemistry of H-terminated Si depends on Si surface orientation, H_2O pH, and the concentration of dissolved O_2 , as reviewed by Henderson [127]. However, outlined here are only the cooperative effects involving H_2O and O_2 , as this may aid in the understanding of gas-phase oxidation.

Ultrapure H_2O (UPW), containing only parts per billion amounts of O_2 , can etch oxide-free Si surfaces. Etching likely occurs via successive hydroxylation of Si atoms until they are dissolved in the form of $Si(OH)_4$ [128,129]. Hydroxide (OH^-) ions probably play a dominant role in the hydroxylation/oxidation process [122], similar to the case of $HF(aq)$ and $NH_4F(aq)$ etching of Si surfaces [121]. For H_2O at room temperature and neutral pH, quantum chemical calculations indicate that an H-terminated Si (100) surface site is at least 1×10^{18} times more likely to react with OH^- than with H_2O [130]. Ultimately, etching by boiling H_2O results in atomically flat H-terminated Si (111) surfaces with triangular etch pits (likely because of

continued oxidation by trace O_2) [128,129] and in substantially rougher H-terminated Si (100) surfaces with pits exposing (111) facets [131]. These differences in morphology have been attributed to preferential OH^- attack of the more polarizable Si atoms of Si-H₂ and Si-H₃ structures at kinks and steps [122,131].

In view of the proposed etching mechanism, oxidation by OH^- should determine the overall etching rate. In the presence of dissolved O_2 , however, the relative reaction rates depend on the relative concentrations of O_2 and of OH^- , and thus on pH [122]. At O_2 concentrations in the ppm regime, the oxidation rate exceeds the etching rate, and hence SiO_2 grows [37,132,133]. The actual mechanism of H_2O -induced oxidation is still unknown. However, there is some evidence for a cooperative (e.g., catalytic) effect between H_2O and the dissolved O_2 . In a study utilizing isotopically marked H_2O , it was shown that at least 85% of the O incorporated into a 5–7 Å thick oxide formed at room temperature originates from H_2O and not from dissolved O_2 [134]. This was taken as an indication that O_2 activates the H-terminated Si surface or acts as a catalyst for H_2O oxidation of Si, rather than oxidizing substantial amounts of Si itself. The possibility of isotopic exchange between $H_2^{18}O$ and $^{16}O_2$ or Si $^{16}O_2$ formed from $^{16}O_2$ was not discussed. Rather, the authors proposed that the role of the dissolved OH^-/O_2 couple is to anodically polarize the electrode, driving the H_2O oxidation reaction [134]. Water-induced oxidation of P-doped n^+ -Si in fact proceeds dramatically faster than for n^- or p^+ -Si, supporting a field-assisted oxidation mechanism [37,132]. However, at doping levels of 1×10^{20} atom/cm³, chemical effects (e.g., catalysis) caused by the dopant atoms also may have to be considered.

A very detailed mechanistic picture of the initial stage of H-terminated Si oxidation by O_2 -containing H_2O has been proposed by Cerofolini et al. [135]. They suggested that H_2O rapidly attacks the low concentration of F impurities on an HF-etched Si, thus forming OH groups. It was speculated that such sites then bond thermal excitons, i.e., electron–hole pairs, leading to Si–Si back-bond cleavage and thus to the formation of surface Si^- and subsurface Si^+ . The subsurface Si^+ then bonds molecular H_2O in the form of a Lewis acid–base adduct, while the surface Si^- ionizes dissolved O_2 and reacts with it, forming $SiOO^-$. Proton transfer from the H_2O adduct to the $SiOO^-$ results in Si–OH (silanol) and Si–OOH [135]. In this largely speculative model, O_2 acts as a necessary ingredient for subsurface hydroxylation, and thus ultimately for the formation of Si–O–Si bridges via silanol condensation. However, direct evidence for these reaction steps is not available.

In conclusion, O_2 -enhanced oxidation appears to occur through a cooperative effect involving H_2O and/or OH^- and O_2 . As discussed below, similar phenomena may be operative in gas-phase oxidation of H-terminated Si. Details of the mechanism are still unclear, however, largely for lack of in situ studies.

9.4.3 Oxygen and Air Interaction With Hydrogen-Terminated Silicon Surfaces

Early experimental studies of H-terminated Si oxidation by O_2 were done in UHV. At temperatures below the onset of recombinative H_2 desorption (e.g., above $\sim 600K$) on H-terminated Si (100) [136], substantial oxidation by typical UHV gas exposures in the Langmuir regime does not occur. Work performed on a number of H-terminated Si surfaces prepared in UHV by atomic H exposure of well-defined Si reconstructions, e.g., for mono- and dihydride H-terminated Si (100), both flat and vicinal, and for H-terminated Si (311) surfaces [136,137] showed that room temperature O_2 exposures in the 1000-L regime did not result in observable OH formation or in the formation of oxide film exceeding an estimated $0.7 \text{ \AA}$ in thickness. There is no observable reactivity enhancement by steps. Dangling bonds appear to be needed for oxidation in this exposure regime.

The initial room temperature sticking coefficient of O_2 on HF-etched Si (100) was determined to be 1×10^{-12} , according to Westermann et al. [138] and Kawamura et al. [139]. Even lower sticking coefficients of HF-etched Si emerged from studies by Morita et al. [37], who detected less than $1 \text{ \AA}$ SiO_2 after 1 week of 4:1 $N_2:O_2$ exposure at 1 Torr ($\sim 10^{14} \text{ L } O_2$); and by Niwano et al. [140], who found that complete surface oxidation, as detected by in situ IR spectroscopy, in 1 Torr O_2 required an exposure of $1 \times 10^{15} \text{ L}$. All these results point to the potential role of impurities, such as H_2O , in the initial surface oxidation of H-terminated surfaces.

As is the case for H_2O oxidation, atomic-scale information on the O_2 oxidation mechanism is scarce. Only Cerofolini et al. speculated on an oxidation scheme by O_2 or dry air [135]. They argued that O_2 molecules dissolved in the water used to rinse HF-etched Si could react with Si^- sites that are negatively polarized through interaction with H_2O . On longer time scales, reaction also is suggested to occur through O_2 ionization by Si^- , which is created by the trapping of a thermally generated excitation. This would explain why the apparent activation energy for oxidation in the initial stage is close to the exciton energy and hence to the Si band gap of $\sim 1 \text{ eV}$.

Photo-induced mechanisms also have been considered. Photon-stimulated H desorption [141] and formation of reactive O radicals in O_2 gas [142] have been identified as relevant H-terminated Si oxidation channels in the presence of UV light. Using intense ultrashort laser pulses, near-IR light or even visible light can accelerate oxidation of H-terminated Si (111) [143], suggesting that electron–hole pairs play a key role in promoting oxidation.

The H-terminated Si reactivity to O_2 is higher at elevated temperatures, which are relevant for ALD growth of gate oxides. A comprehensive investigation by in situ IR spectroscopy and XPS [144,145] reveals that only $\sim 1 \times 10^9 \text{ L}$ of dry O_2 is required to almost completely oxidize atomically flat H/Si (100) (2×1) [prepared by a H_2 anneal of Si (100) (2×1)] and H/Si

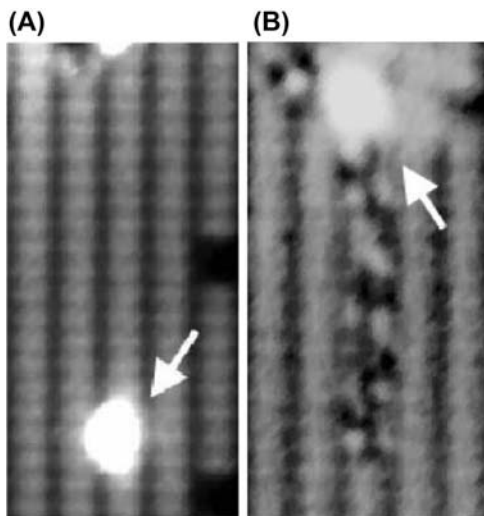


FIGURE 9.4-30 Filled state scanning tunneling microscope images of an H/Si (100) (2×1) with isolated dangling bonds (A) before and (B) after exposure to 10 L of O_2 molecules at 530K, performed at room temperature with a sample bias voltage of -2.0 V and tunneling current of 20 pA. The bright ball-like feature denoted by an arrow is a dangling bond. Note the irregular oxidized region (“wire”) [146]. Reprinted figure with permission from K. Kato, H. Kajiya, S. Heike, T. Hashizume, T. Uda, *Phys. Rev. Lett.* 86 (2001) 2842–2845. Copyright (2001) by the American Physical Society.

(111) (1×1) (prepared from an NH_4F wet etch). On H-terminated Si (111), the growth is predominantly lateral in nature, while on H-terminated Si (100) both lateral and vertical growth occurs and the area density of oxide patches is higher. An in situ microscopic study in UHV revealed some mechanistic insight into how the initial ML oxidation may proceed at elevated temperatures [146]. It appears that even a small concentration of isolated dangling bonds is sufficient to facilitate substantial oxidation, at such sites, oxide stripes as long as 15 dimer units, as shown in Fig. 9.4-30 are observed after O_2 exposures of only 10 L at 530K. In these experiments performed at elevated temperatures, the H-termination is preserved during oxidation, with O insertion into the Si–Si back-bonds [144–146].

First principle calculations (using mostly Density Functional Theory [147]) have addressed the O_2 interaction with H-terminated Si. If residual dangling bonds are present, O_2 preferentially dissociates at these sites and initiates oxidation, which proceeds along dimer rows via H migration [146]. At low temperatures, the theory confirms that only a physisorbed state can be stabilized [147,148]. If a reaction is forced to occur, O insertion into Si–Si

back-bonds is energetically favorable over insertion into Si–H bonds [149], consistent with the observation of Si–H and the absence of any hydroxyl (Si–OH) at the surface.

Given the large O₂ exposures (1×10^9 – 1×10^{15} L) required to oxidize H-terminated Si surfaces, cleanliness issues caused by impurity-mediated reaction are expected to come into play. Notably, air oxidation of H-terminated Si (100) and Si (111) accelerates with increasing humidity [37,132,140,150,151]. However, a two-step mechanism is required to model the time-dependent oxidation data [150–152]: the first step is a slow nucleation, followed by a faster subsurface oxidation next to the attacked sites. The nucleophilic attack of back-bonds polarized by subsurface O may facilitate oxidation, resulting in two-dimensional island growth. The rate constants for H-terminated Si (111) are found to scale with humidity, indicating that H₂O is responsible for initial surface modification. The newly formed hydrophilic sites then promote physisorption of additional H₂O. The dependence on O₂ only becomes important after this initial water-induced oxidation [150,151].

Both for air [150–152] and O₂ oxidation [139] at room temperature, the reaction appears to accelerate as soon as complete incorporation of O into the back-bonds has taken place (labeled H/SiO_x). This observation suggests a reaction mechanism involving physisorbed precursor molecules (O₂ or H₂O) that diffuse across the surface initially made up of H/Si and H/SiO_x areas. These molecules can either desorb or react with the surface. If the desorption barriers are faster for H/Si than for H/SiO_x, then the residence time of the precursor molecules on the H/SiO_x areas leads to faster oxidation as soon as O incorporation is complete.

The role of dopants in the Si substrate also had been shown to influence H-terminated Si oxidation rates. For instance, oxidation of n⁺-Si proceeds much faster than for n[−] or p⁺-Si, indicating that substrate electrons affect reactivity [37,132]. The initial oxidation rate also appears to scale with surface roughness [152], although some authors disagree [140]. This finding was taken as an indication that oxidation starts from steps or defects and continues two-dimensionally (e.g., via Si back-bond polarization and nucleophilic attack) until the first ML is completed [152]. In the absence of defects, for instance, an atomically flat H-terminated Si surface can remain perfectly oxide-free in humid air for at least 15 min [153].

Finally, ambient contaminants other than H₂O, such as organics or radicals, cannot be disregarded. For instance, the presence of organic species has been invoked to account for changes in the electrical surface properties of H-terminated Si [154]. Such species may also affect the oxidation rate. While all air oxidation studies implicitly assume that other contaminants are either absent or did not influence oxidation, experiments utilizing high-purity O₂/H₂O mixtures have shown that this is not the case as described in the next paragraph.

In situ IR measurements have been performed to study O₂ interaction with various H-terminated Si surfaces and to unravel the oxidation mechanisms [155,156]. These studies use high-purity O₂ and focus on comparing

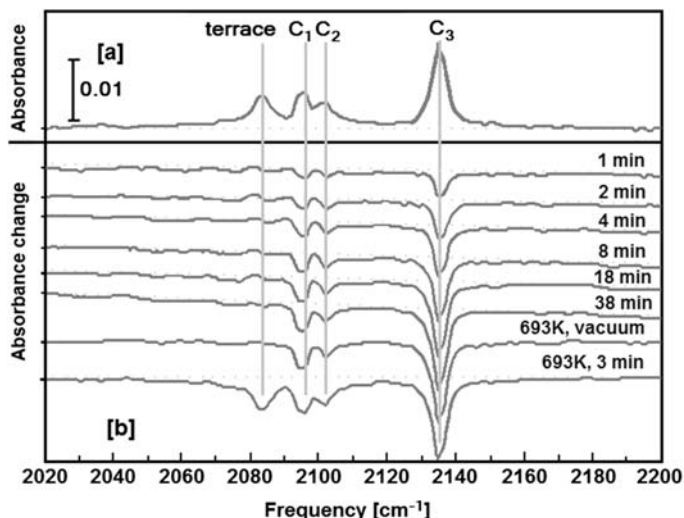


FIGURE 9.4-31 Multiple internal reflection infrared spectra for H-terminated Si (111) 9 degree miscut along $1\bar{1}\bar{2}$ (dihydride steps) exposed to 0.02 mTorr O_2 at 573K, subsequently annealed to 693K, and exposed to O_2 at the same temperature. (A) Absorbance of starting surface; spectral reference: H-terminated Si (111) oxidized by O_2 at 693K. (B) Difference spectra for the subsequent processing steps; the spectrum of the starting surface is used as a spectral reference, i.e., negative absorbance indicates the loss of species from the surface. C_1 , C_2 , and C_3 are modes related to step dihydrides. An exposure of 1 min corresponds to a dose of 9×10^5 L O_2 [158]. Used with permission from Rutgers University.

technologically relevant HF-etched Si (100) to structurally well-defined model surfaces, such as atomically flat H/Si (100) (3×1) prepared in UHV [156] and flat and stepped H-terminated Si (111) prepared by an NH_4F wet etch [51,157]. Structural and chemical information is readily available from the Si–H stretch modes (best probed by MIR) and direct detection of Si–O modes is usually done using transmission geometries.

Fig. 9.4-31A shows the Si–H stretch spectrum of an H-terminated Si (111) surface miscut at 9 degrees in the direction exposing steps terminated by dihydrides (Si–H₂) [158]. Terrace monohydrides (~ 2083 cm^{−1}) are distinguished from the step modes characterized by three distinct vibrations at $C_1 = 2095$, $C_2 = 2102$, and $C_3 = 2136$ cm^{−1} [159]. Fig. 9.4-31B shows that the step modes react much faster than the terrace monohydride upon O_2 exposure at 573K, a temperature typical of ALD growth. In this example, the disappearance of the Si–H stretch modes in the 2000 cm^{−1} range is caused not by the reaction of H with O (to form OH for instance) but by the insertion of O into the Si–Si back-bonds, resulting in a large Si–H stretch blue shift [155]. For instance, as a flat surface is oxidized, modes corresponding to monohydrides with singly, doubly, and triply oxidized Si atoms, observed at 2150, 2200, and 2250 cm^{−1} (not shown), appear [160]. In all cases, there is no

measurable absorption in the $3658\text{--}3750\text{ cm}^{-1}$ range corresponding to hydroxyls attached to Si atoms with varying degrees of oxidation.

Similar kinetic experiments have been performed for a variety of surfaces, including flat H-terminated Si (111) and for H-terminated Si (111) with monohydride or dihydride steps [155,160]. For example, Fig. 9.4-32 shows the oxidation kinetics for monohydrides on Si (111) terraces compared with those of mono- and dihydrides at steps. The oxidation rate is highest for dihydride steps, then monohydride steps, and slowest for terrace monohydrides.

Similarly, the rates of reactions for mono-, di- and trihydrides have been measured for flat (100) and (111) surfaces and for a variety of reconstructions and steps [155,160]. The results can be summarized as follows:

- For comparable substrate orientation, the reaction rates increase according to: terrace monohydride < step monohydride < dihydride < trihydride (up to one order of magnitude).
- On H-terminated Si (111), the reaction rates for terrace hydrides increase according to: flat < stepped with monohydrides < stepped with dihydrides.

Based on the available data [160], some inference could be drawn on the oxidation mechanism:

- Oxidation occurs by direct insertion of O into Si–Si back-bonds, i.e., without H removal or OH formation.

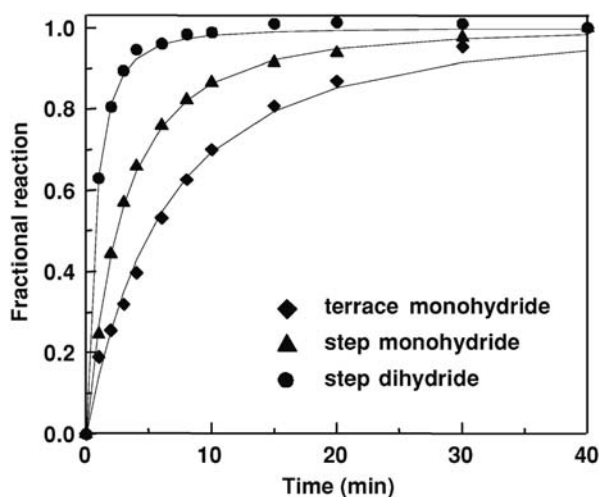


FIGURE 9.4-32 Kinetics of the Si–H peak decay on flat and stepped H-terminated Si (111) surfaces for the terrace H, step monohydride (on $11\bar{2}$ miscut surface), and step dihydride (on $11\bar{2}$ miscut surface) in 1.3-mTorr O_2 at 573K. An exposure of 1 min corresponds to a dose of $6 \times 10^7\text{ L O}_2$ [160]. Reprinted with permission from X. Zhang, Y.J. Chabal, S.B. Christman, E.E. Chaban, E. Garfunkel, *J. Vac. Sci. Technol. A19* (2001) 1725. Copyright 2001, American Institute of Physics.

- The activation energies are approximately 1.6–1.7 eV, lower than typical Si–Si bond energies (e.g., 2.31 eV [161]), indicating that oxidation occurs without the Si–Si bond breaking before O insertion.
- Differences in the reaction rate of mono-, di-, and trihydrides are most likely caused by different activation energies resulting from variations in steric hindrance.
- Variations in reaction rates are largely consistent with differences in accessibility of the back-bonds to O₂ and thus higher reactivity is found for the highly accessible back-bonds of strained tilted dihydrides at H-terminated Si (111) steps and for the back-bonds on rough surfaces in general, when comparing the same type of hydride on different substrates.
- In some cases, a minor effect may be a lowering of the activation energy at strained sites, e.g., for strained mono- and dihydrides on Si (100) (3×1), leading to increased reactivity.
- O-hopping immediately after insertion (dissipating the high free energy of oxidation) is a possible cause for the observed enhancement of terrace oxidation by nearby steps/defects and the absence of the Si–H stretch observed with a single oxygen atom in its back-bond.
- A reaction order of 0.7, obtained by the O₂ pressure dependence, shows that the interaction of O species simultaneously present at the surface lowers the probability for reaction; this might be because of the presence of a layer of temporarily physisorbed O₂ atop the H-terminated Si surface, whose area density scales less than linearly with gas-phase pressure because of O₂–O₂ collisions.

It was further shown that the oxidation mechanism observed in the low-pressure regime is applicable to atmospheric pressures [155,160]. For atmospheric conditions, IR data yield a rate constant of 0.052 min^{-1} , fairly close to a value of 0.029 min^{-1} extrapolated from the kinetic data discussed above, obtained at much lower pressures. This indicates that the same oxidation mechanism applies across five orders of magnitude and a temperature range of 240K. Further extrapolating the kinetics at atmospheric pressure to room temperature, a reactive sticking coefficient of O₂ on H-terminated Si (111) of $\sim 10^{-13}$ is expected [155,160]. This number is identical to room temperature sticking coefficients on H-terminated Si (100) of $\sim 1 \times 10^{-12}$ to 1×10^{-15} obtained in other studies [37,138,140], again indicating that the same oxidation mechanism may apply.

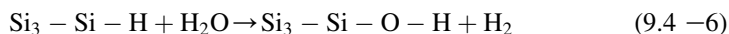
9.4.4 Water Vapor Interaction With Hydrogen-Terminated Silicon Surfaces

The reactivity of UPW at H-terminated Si surfaces is dominated by OH[−]. In the vapor phase, neutral or ionic species formed by H₂O dissociation cannot

play a role because the H₂O photo-induced fragmentation is insignificant (OH reaction probability in the gas phase is $\sim 1 \times 10^{-45}$ at a temperature of 573K as determined by the Boltzmann factor). Molecular O₂, on the other hand, may be present in H₂O vapor, because the liquid H₂O reservoir contains substantial amounts of dissolved O₂ after contact with atmospheric air. Therefore, special care must be taken to reduce the O₂ content in the liquid bath by bubbling with inert gases into it or by other chemical purification methods to remove O₂ from water [129,162].

Tagaki et al. [163] first demonstrated the low reactivity of H-terminated Si exposed to H₂O vapor. In a UHV environment at room temperature, no detectable H₂O adsorption on H/Si (100) (2×1) was observed with exposures in the Langmuir regime (10^{-6} – 10^{-4} Torr-sec range). At low temperatures (e.g., 90K), H₂O physisorbs, forming ice clusters, and desorbs below room temperature without reacting with the substrate. At least 5×10^7 L H₂O must be supplied to H-terminated Si (111) at room temperature [164] to obtain an oxide thickness of about 2 Å. The H termination is preserved during the formation of thin SiO₂ films, similar to the case of O₂ oxidation discussed in the previous paragraph. The generation of activated Si–Si back-bonds was invoked as the rate-limiting step for these relatively lower temperatures. Initial oxidation is extremely slow even at 598K, while above 723K greater than 3-Å of SiO₂ forms rapidly. In this higher temperature regime, the oxidation rate is limited by thermal desorption of the H overlayer from the Si substrate [164]. This is in line with the findings of Kim et al. [165] who reported that a long (also unspecified) H₂O exposure of H-terminated Si (100) in an ALD reactor at up to 573K results in 1–1.5 Å of SiO₂, while at 623K and above much thicker oxide films are formed. They interpret their observations as an indication that, in fact, no reaction should occur below 573K. In the absence of any direct spectroscopic evidence, oxidation at higher temperatures was tentatively attributed to the creation of dangling bonds through H desorption and subsequent OH formation with the rapid insertion of O into Si–Si back-bonds. Zaibi et al. [166], utilizing photoemission yield spectroscopy, also found a reaction threshold at 623K, a temperature at which they indirectly inferred a substantial OH concentration on H-terminated Si (111) already upon H₂O doses as low as 50 L.

A number of theoretical studies have addressed the interaction of H₂O with H-terminated Si. The physisorption energies on H-terminated Si (100) and Si (111) are estimated at -0.16 eV [167] and -0.13 eV [168], respectively. For hydroxylation via:



overall reaction enthalpies of -0.70 [131], -0.75 [169], and -0.59 to -0.69 eV [170] are reported, with corresponding barrier heights of 2.13, 1.60, and 1.85–2.05 eV. There is experimental evidence that trihydride configurations are hydroxylated more easily than dihydride structures [170]. For all

configurations, however, hydroxylation requires substantial thermal energy to occur and is therefore improbable at room temperature.

Recent IR absorption studies of H₂O vapor interaction with H-terminated Si have used a transmission geometry and focused on ensuring a low O₂ concentration by extended bubbling of the H₂O with ultrapure N₂. Such precaution is essential in view of the potential influence of O₂ in H₂O vapor on oxidation. Most experiments are performed under typical ALD conditions (e.g., 573K) in a lab-built ALD reactor [171]. For such spectroscopic experiments, it is convenient to use D₂O instead of H₂O to separate surface reactions from possible fluctuations because of residual H₂O vapor in the spectrometer, while leaving the oxidation chemistry unaffected [171].

The main result illustrated in Fig. 9.4-33 is the stability of H-terminated Si surfaces under very high concentrations of H₂O exposures at relatively high temperatures. Whether the surface is ideally H-terminated, Fig. 9.4-33A, or atomically rough with several types of hydrides, as shown in Fig. 9.4-33B, the Si–H stretch modes remain unchanged after D₂O exposure in the 1×10^8 L regime at 573K [172], as only a few percent of isotopic H–D exchange takes place. On the surface of H₂O-exposed H-terminated Si (111), there is no evidence of H-bonds to oxidized Si atoms (O_n–Si–H, 2130–2300 cm^{−1}) [144], isolated OH groups (Si–OD, 2700–2760 cm^{−1}), or SiO₂ phonon signals (~ 1000 –1200 cm^{−1}) [172]. These results establish that oxidation and hydroxylation of H-terminated Si (111) are insignificant under the reaction conditions studied.

On H₂O-exposed H-terminated Si (100), there is evidence for incorporated O and isolated OD groups on oxidized Si sites (2760 cm^{−1}), as shown in Fig. 9.4-33B. The area density of O and OD species has been quantified by comparison to transmission IR spectra of well-characterized reference samples, as shown in Fig. 9.4-33B. For SiO₂ phonon intensity calibration, a ~ 5.5 Å film prepared by wet-chemical oxidation of H-terminated Si (100) in a H₂SO₄:H₂O₂ solution was used [126]. For OH intensity calibration, a well-defined OH overlayer on Si (100) prepared in UHV was chosen. Comparison shows that at most 0.1 Å of SiO₂ (5×10^{13} O ions cm^{−2}) are formed and that the OD area density amounts to 2% of an ML (1.3×10^{13} OD cm^{−2}) [173].

Comparing the reactivity of structurally and chemically defective H-terminated Si (100) with that of completely passivated, flat H-terminated Si (111), hydroxylation is found to occur only at certain defects present in low concentration, such as oxide or specific step sites. In conclusion, in the 10^8 L exposure regime, well beyond exposures relevant to ALD processes, O₂-depleted H₂O does not oxidize H-terminated Si (100) nor (111) at 573K to any significant extent.

Experiments with even higher D₂O exposures (nearly 1×10^{10} L) were performed on H-terminated Si (111) at 573K (Fig. 9.4-34) [126]. The high degree of surface order makes it possible to discern IR features because of

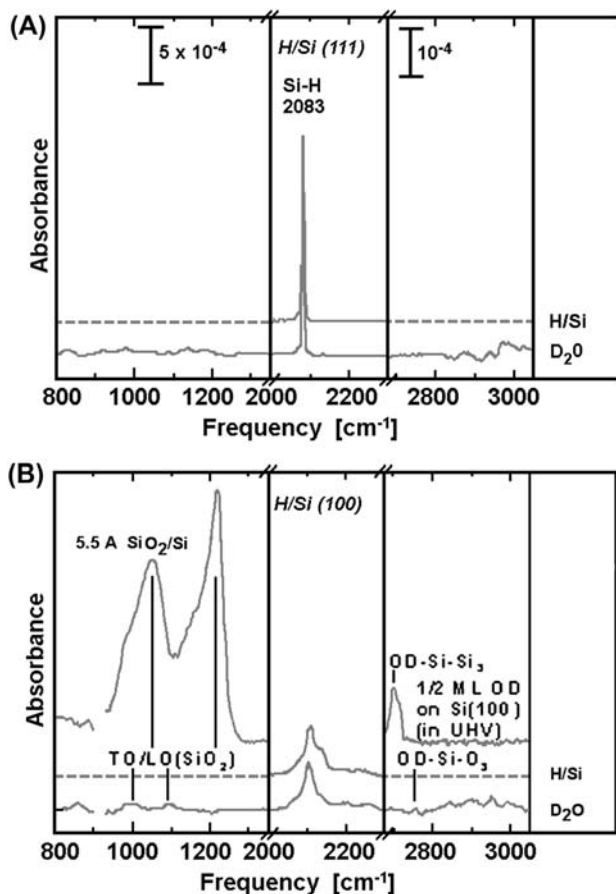


FIGURE 9.4-33 Transmission infrared spectra of H-terminated (A) Si (111) and (B) Si (100) measured before and after exposure to $(1-5) \times 10^8$ L D_2O (10 mTorr) in N_2 carrier gas (1 Torr) at a sample temperature of 573K. Reference spectrum in oxide phonon and O-D stretching regions: H-terminated Si. Note that a signal at $\sim 920 \text{ cm}^{-1}$, because of Si-H bending modes of the H-terminated Si (100) reference surface, has been removed for clarity. Reference spectrum in Si-H stretching region: Si with native oxide. For comparison, we show infrared spectra of well-known oxide and hydroxyl over layers described in the text [126]. *M.M. Frank, Y.J. Chabal, in: A.A. Demkov, A. Navrotsky (Eds.), Materials Fundamentals of Gate Dielectrics, Kluwer Academic Publisher/Springer, Netherlands, 2005, p. 367. Reprinted with kind permission of Springer Science and Business Media.*

different H-Si and D-Si species and therefore estimate or place upper limits on several mechanisms including isotopic exchange. Specifically, the authors found the following features: $<1\%$ ML OD; $68 \pm 3\%$ ML $\text{Si}_3\text{-Si-H}$; $\leq 3\%$ ML $\text{O}_3\text{-Si-H}$; no or very little $\text{O}_{1,2}\text{-Si-H}$; $24 \pm 2\%$ ML $\text{Si}_3\text{-Si-D}$;

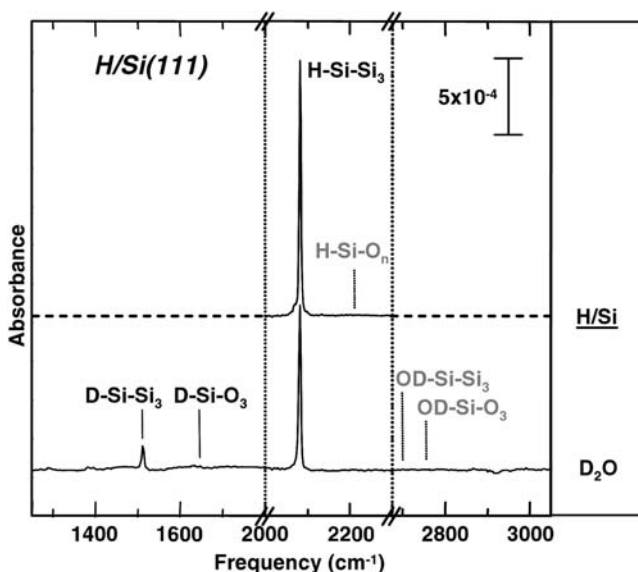


FIGURE 9.4-34 Transmission infrared spectra of H-terminated Si (111) before and after exposure to 7×10^9 L D_2O (10 mbar) in ultrapure N_2 carrier gas (1 bar) at 573K. H-terminated Si is the reference for D—Si and O—D stretching regions and oxidized Si reference for the H—Si stretch region [126]. M.M. Frank, Y.J. Chabal, in: A.A. Demkov, A. Navrotsky (Eds.), *Materials Fundamentals of Gate Dielectrics*, Kluwer Academic Publisher/Springer, Netherlands, 2005, p. 367. Reprinted with kind permission of Springer Science and Business Media.

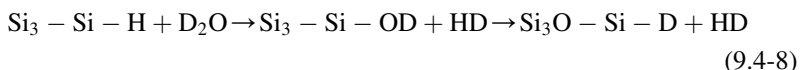
$10 \pm 3\%$ ML $O_3\text{—Si—D}$; and no or very little $O_{1,2}\text{—Si—D}$. Here, 1 ML is defined as the area density of H atoms on the initial H/Si (111) (1×1) surface at $7.83 \times 10^{14} \text{ cm}^{-2}$. The total area density of $\sim 30\%$ ML O ions corresponds to $\sim 0.6 \text{ \AA}$ of SiO_2 formed in the reaction. In fair agreement with this, a small phonon signal (not shown) indicates the presence of $\sim 0.3 \text{ \AA}$ of SiO_2 . These observations yield an upper level for the reactive sticking coefficient of $\sim 1 \times 10^{-10}$. Although most experimental studies have shown that H_2O vapor oxidation of H-terminated Si is a slow process [163–166], this particular work arrives at an even lower reactivity, suggesting that dissolved O_2 was present in higher quantities in the previous experiments [126].

The similar area densities of D and O atoms would be compatible with an initial hydroxylation step:



followed by a rapid OD decay with O insertion into a Si—Si back-bond. Such OH decay on pristine $Si\text{—}Si_3$ sites is known to occur below 600K from studies

on Si (100) surfaces in UHV [174]. The overall initial oxidation mechanism at elevated temperature may therefore be:



with hydroxylation being the rate-limiting step. A preference for hydroxylation over direct H–D exchange also is supported by a density functional study of H₂O reaction pathways with H-terminated Si (100) [169]. The activation barrier and the overall enthalpy of reaction for hydroxylation are found to be nearly 0.7 eV lower than for proton exchange. However, as discussed in detail for O₂ oxidation of H-terminated Si, pathways and barriers may be different for different Si surface orientations. Therefore, comparison of H-terminated Si (100) and Si (111) may be difficult.

For oxidation at 673K, appreciable quantities of stable OD groups are observed once the first SiO₂ ML is completed (not shown) [126]. Such OD groups are bonded in the local arrangement O₃–Si–OD and are thus likely stabilized against decay by the back-bonded O. Because OH groups on pristine Si sites at room temperature are kinetically stable [174], the oxidation mechanism may be fundamentally different at 300K than at elevated temperatures.

In conclusion, in situ IR studies show that H-terminated Si surfaces react faster with H₂O vapor than with O₂ at 573K. Assembling available evidence, the reactivity of H-terminated Si between room temperature and 573K appears to increase in the order: H₂O < O₂ < humid air. This surprising finding may indicate that there is a cooperative effect between H₂O and O₂; such as a two-step oxidation mechanism in which one species opens up the surface while the other is needed for substantial oxidation. Alternatively, the reactivity of ambient air might be because of impurity species not yet considered; for example, hydrocarbons, O₃, or radicals. To come to a conclusive answer, high-purity gas experiments utilizing O₂/H₂O mixtures and impurity addition of O₂ or H₂O will be required, including quantum chemical calculations coupled to these experiments.

9.4.5 Other Types of Contamination of Hydrogen-Terminated Silicon Surfaces

An important contaminant found on the wafer surface after HF etching is F. Fluorine contamination is the most misunderstood of all the contaminants found on HF-etched surfaces. When wafers are treated with concentrated HF without H₂O rinsing, XPS reveals that ML quantities of F are present on the surface [63]. This could lead to the conclusion that F is the fundamental surface termination responsible for the amazing surface passivation [62]. Indeed, F from the HF is identified as playing an important role in reducing the number of defects in gate oxides formed on these surfaces [175], as noted in

Section 9.3.1. Although there is no doubt that F is present, the form in which it is found and how it is bound to the surface are uncertain. An important experimental result by Gräf et al. [70] states that rinsing in H_2O with immersions as short as a few minutes significantly reduces surface F concentrations. Surface F concentrations on Si wafers that are immersed in dilute HF are in the range of a few percent of an ML [69,70], showing that the surface F concentration can vary vastly. The H content on the surface, on the other hand, does not vary substantially with rinsing or with changes in HF concentration in the solution. Thus, it is clear that F must be thought of as a surface contaminant rather than as the fundamental surface termination.

Although it is generally accepted that the H-termination resulting from HF etching explains the hydrophobicity and the passivation of Si surfaces, there is still disagreement as to the nature of the F on the surface. It is clear that some of the F must be physisorbed to the hydride-covered Si surface in chemical forms, such as HF, H_2SiF_6 , or $(\text{NH}_4)_2\text{SiF}_6$, or as ions, such as F^- , HF^{2-} , or SiF_6^{2-} [176]. The reason for postulating physisorbed species is that the concentration of these contaminants changes greatly with rinsing without changing the surface H concentration. Thus, physisorbed fluorides can explain many of the observations of F on Si, in addition to F directly bonded to the surface [38,70,71,175]. The H_2O rinsing experiments of Gräf et al. [70] show that F disappears from the surface at the same time that O is taken up by the surface, indicating that an exchange reaction is occurring. Sunada et al. [38], on the other hand, argue that F-termination prevents oxidation until the F concentration falls below a critical value, after which oxidation can proceed normally. Interesting new data suggest that the F is in fact subsurface and only comes to the surface upon oxidation of that surface [51]. Clearly, the chemistry of F on the Si surface after HF etching is an important issue still being debated and needs to be studied further.

HF-etched Si surfaces can be prepared to be O-free. As mentioned in the previous paragraph, however, O concentrations on the Si surface are found to grow as a result of prolonged H_2O rinsing [38,70]. The atomically flat Si (111) wafers produced by etching in NH_4F , on the other hand, are resistant to oxidation and show negligible amounts of O after H_2O rinses as long as 10 min [75]. The conclusion from this result is that the oxidation rate must somehow be related to the step density on the surface. This may lead to the conclusion that H_2O or OH^- attack the step edges more readily and lead to oxidation; however, the Si boiled in H_2O experiments of Watanabe et al. [85] suggest otherwise. Their surfaces are found to be terminated completely with H, with little if any residual O, demonstrating that step edge attack in H_2O does not necessarily lead to oxide formation. Oxidation appears to be more complex than simple attack by H_2O . Oxidation in liquid solutions has been found to be related to the O_2 concentration in the H_2O used during the rinsing process and the doping type of the wafers [37]. It is interesting to note that lightly doped wafers can be rinsed for 104 min without appreciable oxide growth when H_2O

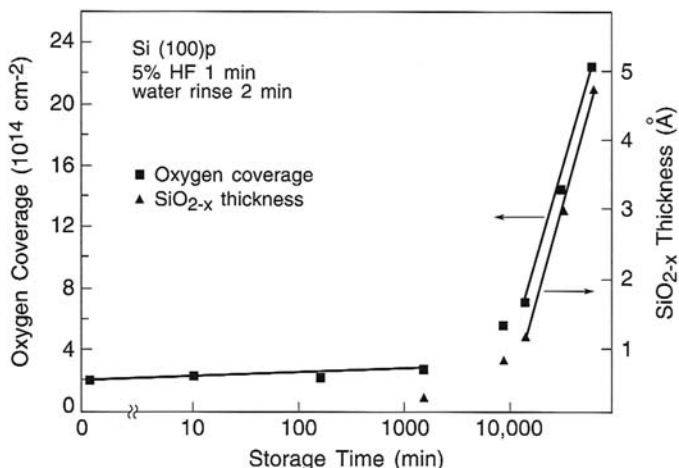


FIGURE 9.4-35 Oxygen coverage (left scale) and oxide thickness (right scale) as a function of storage time in moist air. Oxygen coverage was determined by X-ray photoelectron spectroscopy [180]. Reprinted with permission from D. Gräf, M. Grundner, L. Muhlhoff, M. Dellith, *J. Appl. Phys.* 69 (1991) 7620. Copyright 1991, American Institute of Physics.

with a low O₂ concentration is used. Further research regarding this issue is certainly warranted. Oxidation caused by storage in air is also of interest. Gräf et al. [177] have reported that wafers stored in standard moist air (relative humidity 35–40 vol%), oxidize extremely slowly, growing less than 1 Å of oxide in 7 days (~10,000 min) as shown in Fig. 9.4-35. After 7 days of storage, the oxidation rate increases abruptly, with the oxide thickness approaching 10 Å after 45 days. This bimodal oxidation rate distribution is not understood at this time but has been observed in other work [37,38] and has been postulated to be related to the F coverage on the surface [38]. It should be noted that H-terminated surfaces have been prepared that show less than an ML of O after 7 days of storage both in air [177] and in H₂O [37], given the fact that many researchers still believe that the growth of native oxide begins immediately. The cleanliness of the chemicals available at the time may have influenced the observed oxidation rates, leading to the erroneous conclusion that the oxide growth is an intrinsic rather than extrinsic effect.

Hydrocarbon contamination is a particularly insidious problem for H-terminated HF-treated wafers. Whereas hydrophilic oxide passivated surfaces are relatively resistant to hydrocarbon contamination, H-terminated Si surfaces can be covered with an ML of hydrocarbons extremely quickly if exposed to a contaminated environment. Hydrocarbon contamination can come from the H₂O rinse [71], from contaminants in laboratory air, or from the environments of processing equipment. IR spectra have shown that H-terminated Si samples, prepared in dilute HF and rinsed in O₂-free and high-purity H₂O, then transported through air and loaded into a N₂-purged vacuum

chamber remain hydrocarbon-free, only to be contaminated once the evacuation of the chamber begins. These surfaces are highly sensitive to hydrocarbons and become contaminated unless extreme precautions are taken [69–71]. In fact, clean surfaces have been preserved only in the case where the load locks of the analysis chambers are pumped down a few times to 1×10^{-10} Torr before venting for sample introduction to ensure removal of residual hydrocarbons [75,178]. Part of the problem is that many common contaminants desorb at temperatures above the H-desorption temperature [51]. When this happens, Si carbides are formed and are impossible to remove at reasonable temperatures. In contrast, hydrocarbons on top of oxide-coated surfaces can be made to desorb before the oxide desorbs. Oxide passivation is obviously preferred if the 850–950°C desorption temperatures can be tolerated [49].

Metal contamination is also of crucial importance and is discussed Chapter 2. In brief, wafers etched in HF following the RCA standard clean can be free of metal contamination ($<1 \times 10^8 \text{ cm}^{-2}$) [7]. In other instances, however, Cu^0 is found to be a problem and can be eliminated by adding H_2O_2 to the HF [179]. Copper has been found to be extremely detrimental to the stability of the H-termination by causing increased oxidation rates during H_2O rinses, resulting in 3 Å of oxide growth in just 2 min [180]. This reaction has been attributed to Cu-catalyzed oxidation [180] of the Si, which causes bulk etching and surface roughness upon exposure to contaminated HF solutions. Copper coverage of up to 0.5 ML are obtained on exposure to HF solutions contaminated with only ppm levels of Cu [180]. Copper in HF is known to plate out on semiconductor surfaces. The use of ultrapure HF (and other reagents) in semiconductor processing is, therefore, imperative [1,13].

9.5 STABILITY AND REACTIVITY OF HYDROGEN-TERMINATED SILICON SURFACES IN SOLVENTS

9.5.1 Overview of Solvent Stability

The chemistry of initially hydrogen-terminated Si surfaces greatly affects their reactivity and stability. As discussed in Section 9.4, H-terminated Si surfaces are stable in many environments as long as contaminants are kept at bay. However, the stability of these surfaces in solution is more complex because of solvation effects that can, for instance, result in lower reaction barriers. For instance, while H-terminated Si surfaces are found to be stable in many organic solvents including toluene, mesitylene, and acetonitrile [181,182], these surfaces have been shown to react with different alcohols and recently even HF itself [181,183,184]. Although this is not the focus of the chapter, the stability of surfaces in different solvents and how, by understanding the reactivity of surfaces in these solvents, different functional groups can be attached to the surface and modified *without* oxidation will be briefly

discussed. When immersing the surfaces in solvents, whether for cleaning steps or subsequent functionalization, it is important to note that visible or UV irradiation can greatly affect their reactivity [143,185].

9.5.2 Reaction of Alcohols With Hydrogen-Terminated Silicon Surfaces

Alcohols are important solvents for cleaning and chemical reactions, hence the importance to understand their interaction with H-terminated Si surfaces. For instance, early work has shown that immersion in alcohols leads to the attachment of alkoxy groups, such as methoxy for methanol, through the formation of Si—O—C bonds. This reaction can either be activated thermally [181,186] or photochemically [187–189], which can occur under mild conditions, even at room temperature [190]. In this manner, both short- and long-chain alcohols (e.g., from methanol to decanol) can attach to the Si surface [186,189–191]. In all cases, reactions with alcohols must be carefully controlled as they often result in unintentional interface oxidation [183,187,192,193] and surface pitting [186,194], which are likely caused by trace amounts of water in the alcohol.

In an attempt to overcome this challenge, Michalak et al. [190] immersed H-terminated Si surfaces in anhydrous methanol in a N₂ glove box at room temperature. Using IR spectroscopy, they obtained evidence of the formation of S—OCH₃ modes; however, some concurrent oxidation also is observed despite the use of rigorously manufactured anhydrous solvents to minimize subsurface oxidation [190,195–197].

Later research shows that oxide-free surfaces functionalized with methoxy groups could be prepared by performing this reaction on atomically flat, H-terminated Si (111) surfaces at 70°C [183,198]. Fig. 9.5-36i shows the IR spectra of this reaction where the modes related to the Si—O—CH₃ surface groups are observed without the appearance of modes at ~ 1050 and $\sim 1220\text{ cm}^{-1}$ associated with Si—O—Si bonds, confirming that no subsurface oxidation occurs. The success of this reaction at slightly elevated temperature in a rigorously controlled air- and water-free environment underscores the role of trace contaminants in side reactions; for example, oxidation and the ability to mitigate their effects by optimizing reaction conditions. For Si (111) surfaces, analysis of the IR spectra revealed that a maximum of $\sim 30\%$ of the H atoms are replaced with methoxy groups, suggesting that each methoxy group is surrounded by approximately 6 H atoms [183,198]. Skibinski et al. [184] examined the methoxylated surfaces with STM and XPS; however, they were unable to observe such an ordered structure. The mechanism of this reaction has recently been studied by investigating the effect of an oxidant and visible light on the kinetics of the reaction [199].

Independent of the detailed surface structure, the ability to controllably functionalize the surface opens up the possibility for further surface

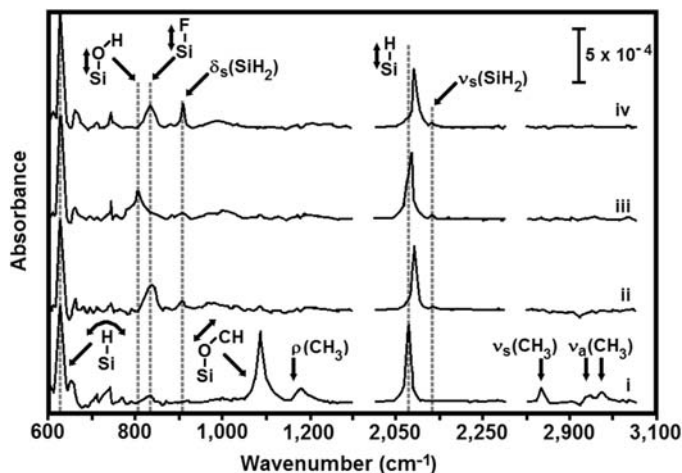


FIGURE 9.5-36 Infrared spectra of surfaces after (i) initial methanol treatment leading to $\sim 30\%$ Si-OCH₃ and 70% Si-H termination, (ii) sequential HF treatment to form the $\sim 30\%$ Si-F termination, (iii) water treatment to obtain $\sim 30\%$ Si-OH coverage, and (iv) the $\sim 30\%$ Si-F surface again. The transmission spectra are referenced to those of the native oxide termination in the spectral regions of 600–700 and 2000–2300 cm⁻¹ and to the H-Si (111) surface in the spectral regions of 700–1300 and 2800–3050 cm⁻¹. Reprinted by permission from Macmillan Publishers Ltd., D.J. Michalak, S.R. Amy, D. Aureau, M. Dai, A. Esteve, Y.J. Chabal, *Nat. Mater.* 9 (3) (2010) 266, copyright (2010).

modifications and selective chemistry where the methoxy sites can serve as templates or active sites for chemical reactions to take place as described below [198]. On the other hand, the fact that alcohols can react with H-terminated Si surfaces, albeit slowly, suggests that the use of alcohols as solvents, especially methanol, should be minimized during other chemical functionalization procedures.

9.5.3 Fluorine Passivation of Hydrogen-Terminated Silicon Surfaces

As discussed in Section 9.3.2, the mechanism of HF etching results in an H-terminated Si surface that is atomically flat and considered chemically inert. As seen in the previous section, the surface can be activated by immersion in alcohols and these surface sites can serve as templates for further chemical reactions. Michalak et al. [198] shows that immersion of the methoxy-passivated Si (111) surface in 49% HF results in the formation of Si-F bonds on the surface as seen by the appearance of an IR mode corresponding to Si-F at ~ 840 cm⁻¹ (Fig. 9.5-36 ii). Additionally, all modes associated with the Si-OCH₃ groups disappear while the Si-H mode blue shifts from 2079 to 2092 cm⁻¹ as a result of interactions with fluorine. It is surprising that the

Si–F bond can be stabilized on the surface because it is highly polar; the origin stems from the perfect nature of the atomically flat Si (111) surface resulting from methoxylation without surface oxidation.

Until recently, the above two-step reaction was the only method to prepare a Si–F termination through wet chemical techniques. However, Skibinski et al. [184], found that the direct immersion of H-terminated Si (111) surfaces in HF results in up to 11% of an ML of Si–F as confirmed via both XPS and STM. This method produces a surface that is randomly functionalized, but the fluorination does not appear to roughen or etch the surface. At this time, the mechanism of fluorination is not known, although it probably does not involve etching or auto-ionization.

In both methods, the stability of the resulting Si–F surface is surprising because of the polar nature of the bond. Nonetheless, the ability to prepare Si–F functional groups via wet chemical methods is beneficial for further chemical modifications [200,201].

9.5.4 Reaction of Water With Fluorine-Terminated Silicon Surfaces

Controllable hydroxylation of oxide-free Si surfaces through wet chemical approaches would increase the variety of chemistry available to functionalize Si by adding reactions optimized for silica surfaces. When perfectly passivated with hydrogen, silicon surfaces are resistant to reaction and/or oxidation in water. However, isolated surface Si–F groups lead to the formation of OH groups on the surface after immersion in water, however, it should be noted, without surface oxidation [200]. Although some hydroxylation is observed after water immersion of a fully F-terminated surface prepared via F₂ deposition in UHV [201], few approaches have proven successful especially in ambient conditions. Therefore, fluorination of H-terminated Si surfaces, either through a two-step reaction [198] or direct immersion in HF [184], is an effective method to directly prepare OH-terminated Si surfaces via wet chemistry techniques. Michalak et al. [198] first demonstrated the selective hydroxylation of the Si–F bond via IR spectroscopy (Fig. 9.5-36 iii). They found that immersion in water resulted in a redshift of the Si–H mode back to its original position and the appearance of the Si–OH stretch mode. Additionally, the change in surface functionality increased the hydrophobicity of the surface as compared with the Si–H or Si–F termination and occurred without oxidation of the surface as seen via XPS. They also found that this reaction was reversible and Si–F termination could be regained upon HF immersion (Fig. 9.5-36 iv). Since then, these OH-terminated oxide-free Si surfaces have been used to explore surface chemistry in exquisite detail [202,203].

9.6 SUMMARY

Two distinct types of surface cleaning processes for Si wafers have been discussed in this chapter. The first leaves a thin chemical oxide behind and the second results in an H-terminated Si surface. Both techniques are extremely effective at cleaning and passivating the Si surface. Three major issues were discussed for each of these classes of cleaning: chemical composition, structure and morphology, and contamination issues.

The composition of chemically grown oxides on Si surfaces is similar to that of the interfacial transition regions of thermally grown oxides. The oxides are largely composed of SiO_2 containing a fraction of sub-oxide species, dominated by Si^{2+} . The largest intrinsic contaminant found on such surfaces is H-bonded OH and H_2O . Minor levels of H in the form of Si—H also are observed. The structure and morphology of CMP Si wafers approaches atomic perfection with a surface roughness of 2 Å RMS. The SC-1 clean is found to increase levels of surface roughness, whereas the SPM and SC-2 cleans leave surface topographies unchanged. The oxide interfacial structure is found to vary with preparation technique; researchers have found that the chemically grown oxide interfaces are substantially rougher than their thermal oxide counterparts. Extrinsic contaminants, such as metals and hydrocarbons, are efficiently removed using RCA cleaning, but there is an increasing concern that the $1 \times 10^8 \text{ cm}^{-2}$ level achieved may still not be sufficiently low for future IC manufacturing. Other areas where future work is necessary include gaining an understanding of how contaminants in the cleaning solutions influence etching and surface topography; elucidating the mechanism of chemical oxidation in the various solutions employed; and learning, finally, how to control the surface structure and composition of oxide-terminated Si surfaces.

Hydrogen-terminated Si surfaces are composed of H atoms covalently bonded to the substrate wafer. The electronic perfection of these interfaces is unsurpassed, indicating an extremely low dangling bond density. Fluorine is now understood to be a minor constituent of the surface and is one of the major intrinsic contaminants. The mechanism of H-termination involves back-bond attack by HF molecules that is facilitated by the charge transfer by Si bonding to highly electronegative elements, such as F or O. Si (100) and Si (111) surfaces are found to be atomically rough when etched in concentrated or dilute HF. This result was originally inferred by the coexistence of many types of Si-hydrides on the surface and was later confirmed using a variety of techniques. High-pH HF solutions lead to etching that can produce rough Si (100) surfaces when the gas products are not carefully removed and the pH accurately controlled. The situation is simpler for Si (111) surfaces that tend to be atomically flatten in high pH HF. These ideally monohydride-terminated Si

(111) surfaces have been characterized by a variety of surface science techniques demonstrating that the high surface quality can be achieved using solution chemistry alone. Such surfaces allow highly detailed studies, which can be used to elucidate the fundamental mechanisms of etching. In this regard, OH^- has been identified as being responsible for the anisotropic etching observed. Surprisingly, H_2O rinsing alone also can be used to achieve a certain level of atomic perfection on Si (111) surfaces, which leads to the conclusion that anisotropic etching can occur during rinsing. Hydrogen-terminated Si surfaces are particularly susceptible to hydrocarbon contamination and extreme precautions must be taken in order for this contaminant to be avoided. Hydrogen-terminated surfaces have been prepared, which are essentially metal free ($<1 \times 10^8 \text{ cm}^{-2}$), but solutions contaminated with Cu must be avoided when using HF. Recent findings indicate that the growth of native oxide occurs much more slowly than previously believed. Researchers found that Si wafers remain essentially oxide free for up to 7 days in both air and H_2O . Future work should be focused on understanding the role of F on these surfaces and the chemistry of HF and OH^- in these solutions. Understanding the role of metal contaminants in HF solutions is also of critical importance. Learning to control Si (100) surface chemistries and morphologies at a level similar to that achieved for Si (111) is of greatest importance.

The foundation of the advances made in the IC industry's understanding of these complex surfaces comes from a number of techniques, such as LEED, IR absorption spectroscopy, STM, AFM, high-resolution electron energy-loss spectroscopy, and high-resolution XPS. Furthermore, theoretical techniques, such as first principles molecular orbital calculations of chemical activation barriers have provided a quantitative basis for understanding important surface chemical reactions. In this chapter, not only is the technology of wet chemical cleaning stressed but also the emphasis on the science behind the wet chemistry. This area of research is growing rapidly and the IC industry will benefit from better understanding that will be central to successful semiconductor processing in the future.

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